

A Framework of Resilience Indicators for Electric Vehicle Emergency Preparedness

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Abstract—As electric vehicles (EVs) become more widespread, their role in disaster resilience and emergency evacuations is increasingly relevant. However, most existing evacuation and resilience planning frameworks lack structured guidance on how to integrate EVs into these systems effectively. This paper reviews the current state of the deployment of EV evacuation plans during disasters, highlighting critical limitations including range anxiety, insufficient charging infrastructure, slow charging rates, and inequitable access across vulnerable communities. At the same time, emerging opportunities such as vehicle-to-grid (V2G) integration, backup power via EVs, microgrid connectivity, and use of EV fleets in evacuation support point to new frontiers of resilience. In response to the lack of a unified evaluation tool, a structured framework is proposed, consisting of 36 resilience indicators covering multiple domains. Namely, equity, charging logistics, mobile backup power, transportation routing, exposure analysis, and community engagement. These indicators are grounded in literature, simulation results, and policy insights, and are categorized by disaster phases (preparedness, response, and recovery). This framework enables communities and policy makers to assess gaps, simulate realistic scenarios, and develop proactive EV evacuation strategies holistically and based on evidence.

I. INTRODUCTION

In recent years, the frequency and severity of climate-induced disasters have dramatically increased, placing unprecedented stress on power systems and evacuation infrastructure [1]. As the transition to electrified transportation accelerates, there is an urgent need to understand how EVs respond under extreme conditions such as wildfires, hurricanes, or grid outages. Although EVs promise environmental benefits, it also introduces new constraints on emergency planning; limited range, long charging times, and insufficient charging infrastructure can all hinder evacuation efforts [2], [3], [4].

This challenge is especially pressing in California, which aims to reach 100% Zero-Emission Vehicle (ZEV) and EV sales by 2035 [5]. While this policy supports climate goals, it also raises concerns about whether current evacuation strategies, largely designed for gas-powered vehicles, remain viable. The state has already experienced an increase in large-scale wildfires and other climate-related hazards [6]; raising the stakes for communities that may rely on EVs during future emergencies. As highlighted in recent literature, EV evacuations are particularly vulnerable to grid failures, congestion at charging stations, and emergency protocols that were designed for gas-powered vehicles and do not account for EV-specific constraints such as charging needs and limited range [7], [8], [9].

Researchers have begun to explore both the risks and opportunities associated with EV-based evacuations. Challenges include access to charging, real-time information, and coordination between public and private stakeholders [7], [9]. However, EVs also offer unique opportunities for resilience through vehicle-to-everything (V2X) capabilities, mobile charging support, and integration with microgrids [8].

Despite growing awareness, existing resilience frameworks often lack specificity when applied to EV evacuations. Approaches used for broader disaster resilience, such as those focusing on hurricanes or wildfires, rarely provide detailed guidance on how to adapt infrastructure, policy, and communication protocols to the specific constraints of EVs [10], [11]. While global frameworks like the United Nations Sustainable Development Goals (SDGs) address resilience and equity at a high level [12], they do not offer actionable guidance tailored to the operational realities of EVs [13].

In response, this paper introduces a structured framework of 36 indicators specifically designed to assess community readiness for EV-based evacuations. These indicators are organized across six domains: equity, exposure, community engagement, charging logistics, mobile and backup power, and transportation routing. Together, they aim to help local governments, utilities, and emergency planners identify vulnerabilities, prioritize investments, and develop more inclusive, robust evacuation strategies. Although this paper mentions hydrogen vehicles, the vast majority of the indicator content focuses specifically on EVs.

Resilience, in the context of this framework, refers to the ability of EV-dependent communities, their supporting charging infrastructure, and interconnected power systems to anticipate, withstand, adapt to, and recover from disaster-induced disruptions, ensuring continuity of emergency evacuation and mobility for all populations. Resilience in this framework spans four interconnected dimensions:

- **Infrastructure resilience:** the operational continuity of EVCS and power systems under stress.
- **Mobility resilience:** the ability of EV users to complete evacuation trips despite range, charging, or routing constraints.
- **Social resilience:** equitable access to EV evacuation resources across vulnerable populations.
- **Institutional resilience:** the coordination capacity among agencies, utilities, and communities before, during, and after disasters.

Moreover, in this framework, the term “community” refers to the collective of individuals, households, organizations, and

institutions operating within a defined geographic region that are involved in or affected by EV-based evacuation planning. This includes, but is not limited to EV owners and users, vulnerable populations, seniors, local government and emergency management agencies, private sector stakeholders, community organizations.

The indicators presented here are grounded in academic literature [2], [4], [9], policy documents [14], [15], and post-disaster infrastructure performance data [16], [1]. Building on the multi-dimensional resilience framework proposed by Assarkhaniki et al. [17], which reviewed resilience literature from 1970 to the present day and identified five Key Dimensions (KDs), social, economic, institutional, infrastructural, and environmental, through a similarity matrix analysis of 38 publications, the proposed framework adopts their distinction between Frequently Addressed Dimensions (FADs) and Peripheral Dimensions (PDs). While FADs such as charging infrastructure and transportation routing reflect core technical capabilities directly analogous to infrastructural and institutional resilience, PDs such as equity and community engagement capture the social and governance dimensions that are critical to inclusive EV evacuation planning but are often underrepresented in technical resilience frameworks.

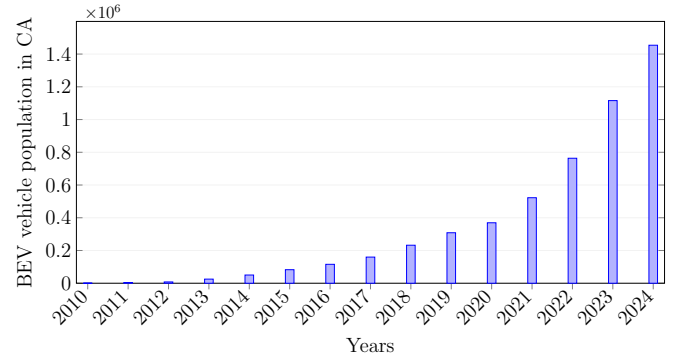
Finally, it is discussed how this framework can support scoring methods such as the Analytical Hierarchy Process (AHP) to assess readiness at the regional level [18], [19] for future work. This creates opportunities for comparison, policy benchmarking, and targeted action to enhance community resilience in an electrified future.

II. SCIENTIFIC LITERATURE REVIEW

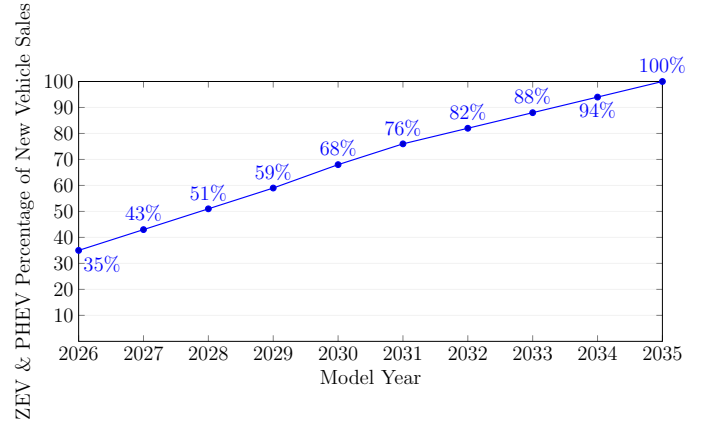
A. Background

California is on track to achieve 100% ZEV and EV sales by 2035 [5]. Figure 1a illustrates the historical growth of the battery electric vehicle (BEV) market in the state from 2010 to 2024. BEV market share in California increased significantly; from approximately 1% in 2013 to nearly 25% in 2024 [20]. In parallel, Figure 1b presents the projected trajectory of ZEV and plug-in hybrid electric vehicle (PHEV) sales, in alignment with California's regulatory objectives [5].

Based on these advancement in EV technology and widespread adoption of clean energy transportation, it has come into consideration that old evacuation plans designed for gas-powered vehicles might become outdated very soon. One major difference between evacuation plans developed for internal combustion engine (ICE) cars and evacuation plans developed for EVs would be the need for grid resilience, deployment of backup chargers, and specific routing for EVs. ICE cars can be refueled in less than 5 minutes at any gas station and can have range of hundred of miles. On the contrary, EVs need charging stations that have power from either grid or any other source and need several minutes or sometimes hours, based on the type of charger available, to have a range comparable to ICE vehicle. These differences forced researchers to investigate different scenarios where EVs would cause a bottle neck and would need specific preparation before any hazard to make sure evacuation from a danger



(a) Battery Electric Vehicle population in California [20].



(b) Projected share of ZEV and PHEV new vehicle sales in California (2026–2035) [5].

Figure 1: California BEV Adoption (2010–2024) and ZEV Sales Requirements (2026–2035).

zone is successful. Moreover, where EVs seem to be a bad option based on above discussion, they can prove to help the grid stabilise after the hazard has occurred i.e. in post-disaster temporal stage. EVs can act as portable battery storage. EVs that have evacuated danger zone can discharge using V2G to maintain the grid so that it can provide power to EVs still stuck in danger zones. This capability is not available for ICE vehicles and number of researchers around the world have carried out simulations and developed solutions to help understand how it would work in case of a natural hazard. Upcoming section talks about literature that has tried to solve this problem using various methods.

Based on previous discussion, existing literature on EV deployment and disaster resilience will be reviewed further to identify critical gaps and inform the development of a more integrated indicator-based framework. Based on the pros and cons of ICE vs. EVs, researchers have increasingly focused on exploring the challenges and opportunities associated with the rapid adoption of EVs. This transformation reshapes mobility patterns and imposes growing pressure on the power grid, requiring expanded electricity generation and distribution capacity [5], [21]. Historically, transportation energy needs were met through a mix of fossil fuels, gasoline, and electricity. However, with a growing shift toward full electrification, the electrical infrastructure faces unprecedented stress [22]. Any

disruption, whether due to natural disasters or technical failures, can cascade into broad impacts, disrupting transportation, charging access, food supply chains, and emergency services. Ensuring a resilient and robust power system thus becomes critical [10], [23].

Several studies have evaluated the vulnerabilities and opportunities introduced by EVs in disaster scenarios. Babaei and Wong [2] provided a comprehensive review of how EVs interact with resilience frameworks, identifying key gaps in current infrastructure planning which are having back-up power available to be deployed, having transportation and evacuation routes developed for EVs. Similarly, Farzan et al. [4] emphasized the need for integrating distributed energy storage with charging infrastructure to ensure operability of EVs, as well as grid during evacuation events. These references provide a concrete reason to focus on transportation and back-up constraints. Donaldson et al. [8] analyzed wildfire events in California, revealing how increased EV loads could strain the grid during peak emergency response times if not properly coordinated, hinting towards better planning before hand as not to cause panic charging of EVs, putting the grid under even more stress.

On the other hand, EVs also offer promising opportunities to enhance resilience. Hussain et al. [9] discussed how EVs, when paired with V2G technologies, can provide backup power and load balancing. Donaldson et al. [7] introduced a model where EV-based evacuations were directly integrated into power system resilience assessments. In addition, Hussain et al. [24] reviewed the cyber-physical vulnerabilities of power infrastructures supporting EV systems, underscoring the need for secure and resilient grid architectures, especially under extreme conditions. Their framework demonstrated that EV mobility can be harnessed to support grid reliability, particularly during phased evacuations. Similarly, Du et al. [25] examined the post-disaster restoration process of urban power networks, showing how EVs, through managed charging strategies, can either accelerate or hinder recovery. All the literature discussed in this paragraph hints toward better planning and development of V2G capabilities.

Yusuf and Summers [3] highlighted logistical challenges posed by EV evacuations, including range anxiety, insufficient charging stations, and infrastructure disparities. Feng et al. [26] raised similar concerns around delays and congestion due to slower charging times compared to conventional refueling. Guo et al. [27] examined the optimal siting of charging stations by explicitly considering drivers' range anxiety, an issue amplified during emergency evacuations.

Beyond power and transport systems, some studies explored cross-sectoral impacts. Saleem et al. [18] proposed a ZEV Evacuation Readiness Score that integrates infrastructure, accessibility, and social vulnerability indicators. Assarkhaniki et al. [17] offered a multi-dimensional framework for resilience that includes technical, institutional, and community-based capacities. These were one of initial efforts to develop a structured framework based on the literature discussed.

Global frameworks further contextualize these findings. The UN Sustainable Development Goals (SDGs) emphasize resilient infrastructure as a foundation for sustainable cities [12].

Finland has operationalized similar indicators into national planning through its Sustainable Development Report [19], [28], as illustrated in Figure 2, which demonstrates how SDG indicators can be structured into a dashboard format for national-level resilience tracking. While these indicators do not focus specifically on EV evacuation, they provide a broader blueprint for resilience assessment and how same structured framework can be developed for only EVs.



Figure 2: Dashboard example for Finland, the top-ranked country on the SDG Index [19].

Overall, the literature underscores a dual narrative: EVs introduce new stressors to existing systems while also offering opportunities for resilience. Bridging this duality requires cross-disciplinary strategies combining energy systems engineering, transportation logistics, disaster management, and public policy.

B. Framework Expansion and Domain Definition

This research provides a novel framework to develop resilience for different regions under stress spanning different temporal regions. The proposed framework builds on the domains proposed in [18], namely "Exposure Analysis", "Infrastructure Vulnerability Analysis" and "Adaptive Capacity of Infrastructure". The proposed framework incorporates these indicators while expanding these to community level indicators as well. The improved framework shifts the focus from individual region evaluation toward developing a comprehensive and systematic framework using the basis from framework presented in [17]. The core novelty lies in structuring the resilience assessment around three temporal stages—*pre-event*, *during-event*, and *post-event*—each representing a critical phase of disaster preparedness and response. Previously presented literature mentions all the simulations and optimization problems developed to help solve these evacuation problems for different temporal stages, but a framework based on these findings has never been developed that can be used by policy developing agencies and communities to build resilience before the disaster, better evacuation conditions during evacuation using EVs and faster recovery of grid after the disaster.

Through an extensive review of prior literature, six domains and 36 indicators presented in Table II have been identified that collectively encompass scoping out EV evacuation readiness around these temporal stages. These domains serve as the foundation for evaluating EV evacuation readiness in a structured manner. Future aim would be to develop a scoring methodology that would help develop a score that can quantify the evaluation.

The six domains defined in this study are:

- Exposure (EX)
- Community Engagement (CE)
- Charging Infrastructure (CI)
- Mobile and Backup Power (MB)
- Transportation and Routing (TR)
- Equity (EQ)

Each domain encompasses a set of indicators that capture its operational characteristics and resilience attributes. These indicators are designed to be measurable, actionable, and reflective of best practices recommended across the reviewed literature. Below is a summary definition of indicators grouped by their respective domains:

- **Exposure:** Indicators assessing geographic and environmental risk factors, such as proximity to hazard zones and historic disaster frequencies.
- **Community Engagement:** Readiness of the community in planning, awareness, and coordination of evacuations with public-private stakeholders during evacuations.
- **Charging Infrastructure:** Availability, distribution, and operational protocols for EV charging under normal and emergency conditions.
- **Mobile and Backup Power:** Use of mobile charging units, V2X support, and backup power sources during outages or evacuations.
- **Transportation and Routing:** Evaluation of evacuation route readiness, real-time navigation support, and responder preparedness specific to EVs.
- **Equity:** Measures related to access, inclusion, and fairness in EV evacuation planning, especially for vulnerable populations.

Among these domains, Exposure appears to be the only domain generally unaffected by the mode of transportation. In contrast, both Equity and Community Engagement are closely tied to the deployment and accessibility of EVs. For instance, Equity encompasses access to EV charging infrastructure [9], while Community engagement involves proactive planning efforts and the establishment of coordination mechanisms among key stakeholders (e.g., planning authorities, utility operators, and emergency responders). These efforts often include formal agreements, such as Memorandums of Understanding (MOUs), as recognized in connected vehicle planning frameworks [29]. Although these domains have been widely studied in the context of disaster preparedness, their relationship with EV-specific factors remains a key consideration in this work.

In the same manner, Charging Infrastructure, Mobile and Backup Power, and Transportation and Routing are uniquely tailored to EVs due to their reliance on electrical energy and grid functionality, unlike internal combustion engine (ICE) vehicles which can refuel quickly and operate independently of electricity. During disasters, disruptions to EV charging infrastructure can significantly delay evacuations and increase risks, especially for vulnerable populations [30]. These domains also require planning for EVCS availability along evacuation routes, a constraint not faced by ICE vehicles. Therefore, the indicators proposed are specifically designed to address the operational limitations and infrastructure dependencies that make EV-based evacuation planning fundamentally distinct.

The next section discusses each domain scientifically and provides references for the use of each domain in the context of resilience of power system and EV friendly evacuation. This section was supposed to present the basic domains that completely cater to making EV usage resilient to any natural hazard, thus making the evacuation planning stronger for any region/country.

III. FRAMEWORK OF RESILIENCE INDICATORS

A basic review of the scientific literature to define the six domains was presented in Section II-B. This section provides a comprehensive review of each domain separately, detailing the key indicators identified for evaluating community preparedness for EV-based evacuations. The proposed framework functions as a unified evaluation tool by organizing 36 indicators across six interrelated domains, Equity (EQ), Exposure Analysis (EX), Community Engagement (CE), Charging Infrastructure (CI), Mobile and Backup Power (MB), and Transportation and Routing (TR), each mapped to specific disaster phases (preparedness, response, and recovery) as represented in Table I. Together, these domains provide a structured, multi-dimensional assessment of a community's EV evacuation readiness. A community or agency can apply this framework by scoring each indicator independently, identifying gaps within and across domains, and prioritizing interventions based on their disaster phase and regional context. The interdependencies between indicators, analyzed through Jaccard similarity, further ensure that the framework captures both domain-specific and cross-domain resilience attributes, forming a cohesive and actionable evaluation system rather than a disconnected checklist. The six domains collectively define the structural foundation of the proposed framework, each addressing a distinct aspect of EV emergency preparedness and community resilience. The complete framework architecture is presented in Figure 3. It should be noted that indicator scoring, evaluation, and result classification are not addressed in this paper, as these constitute a separate empirical undertaking. The framework structure is presented here to provide readers with the broader context and to underscore the significance of a unified, multi-dimensional approach to EV evacuation preparedness.

Each indicator is characterized by its type of assessment:

- Boolean indicators reflect the presence or absence of a specific capability or documented measure (e.g., emergency plans, protocols, or agreements);
- Range (0–1) indicators represent normalized quantitative metrics such as coverage ratios or participation rates;
- Categorical indicators rely on qualitative judgments expressed through ordered levels of performance (e.g., Bad, Medium, Good, Excellent). These levels are determined based on the degree of implementation or availability of a given capability within a community or region, assessed against best practices identified in the reviewed literature. Specifically, the four levels are defined as follows:
 - **Bad:** No plans, infrastructure, or protocols exist for the indicator in question, and no efforts are underway to address the gap.

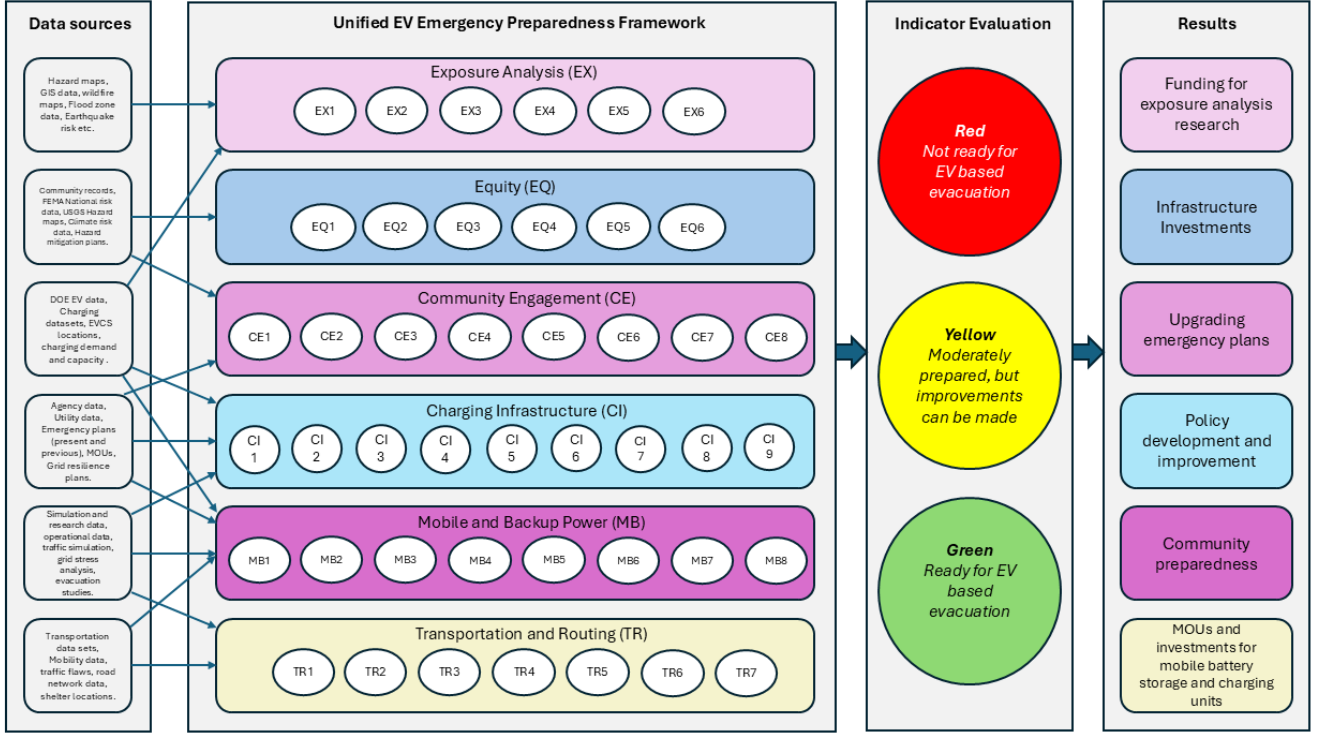


Figure 3: The Unified EV Emergency Preparedness Framework illustrating the flow from domain-specific data sources through six indicator dimensions – Exposure Analysis, Equity, Community Engagement, Charging Infrastructure, Mobile and Backup Power, and Transportation and Routing — to a three-tier readiness evaluation (Red/Yellow/Green) and corresponding actionable outcomes.

- **Medium:** Partial plans or infrastructure exist, but coverage is limited, inconsistent, or not yet operationalized across the region.
- **Good:** Plans, infrastructure, or protocols are in place and functional, but may lack full coverage, regular updates, or integration with other domains.
- **Excellent:** Comprehensive, regularly updated, and fully operationalised plans, infrastructure, or protocols exist, with demonstrated effectiveness and cross-domain integration.

These levels are intentionally kept broad to allow applicability across diverse regional contexts, while still providing a structured basis for comparison and gap identification. A more granular scoring rubric tailored to specific indicators is identified as a direction for future work, particularly in the context of weighted scoring using Multi-Criteria Decision Analysis [18].

Table I: Framework Overview: Domains, Number of Indicators, and Disaster Phase Alignment

Domain	Indicators	Disaster Phase
Equity (EQ)	6	Preparedness
Exposure Analysis (EX)	6	Preparedness
Community Engagement (CE)	8	Preparedness/Recovery
Charging Infrastructure (CI)	9	Response
Mobile & Backup Power (MB)	8	Response/Recovery
Transportation & Routing (TR)	7	Response
Total	36	All Phases

This classification helps communities and practitioners self-assess their level of preparedness and provides a clear framework for developing consistent and comparable evaluation grids across different regions.

Table II summarizes all 36 indicators, their domain categories, abbreviations, supporting references, and indicator types.

A. Community Engagement

As defined in Section I, the term *community* in this framework encompasses five groups: EV owners and users, vulnerable populations (including seniors, people with disabilities, and non-English speakers), local government and emergency management agencies, private sector stakeholders (utilities and charging network operators), and community organizations such as neighborhood groups and non-profits. It is important to note that community engagement in this framework is not limited to actions taken during an active evacuation. Rather, engagement is primarily a *pre-event* activity, focused on building the knowledge, relationships, and coordination mechanisms that enable effective response when a disaster strikes. For instance, EV users can participate in evacuation drills and training programs (CE5, CE7) before a disaster to understand charging needs and safe routes. Local agencies can establish public-private agreements (CE2) and inter-agency coordination protocols (CE4) in advance, so that these mechanisms are

Table II: References used for each indicator

Category	Indicator	Abbrev.	References	Indicator Type
Equity	Emergency Support Programs for Low-Income EV Owners	EQ1	[31], [32]	Boolean
	Participation of Vulnerable Groups in Planning	EQ2	[17], [33], [34], [35]	Categorical
	Mobile Charger Deployment in Underserved Areas	EQ3	[3], [36], [37]	Categorical
	EVCS Access in Disadvantaged Communities	EQ4	[38], [39], [40], [41]	Range (0–1)
	Access to EV Alerts and Multilingual Communication	EQ5	[42], [43], [44], [45]	Categorical
	Subsidized Access to EV Car-Sharing or Loaner Programs	EQ6	[46], [47], [48], [49], [50], [51]	Boolean
Community Engagement	EV User Communication and Readiness during Emergencies	CE1	[52], [53]	Categorical
	Emergency EV Charging Access Agreements and Public-Private Coordination	CE2	[54], [55]	Boolean
	Community Planning for EV-Based Emergency Evacuations	CE3	[2], [56], [18]	Categorical
	Coordination among agencies to Conduct Emergency Response of EVs-Based Evacuation	CE4	[52], [36]	Categorical
	Community Training and Awareness Programs	CE5	[57], [58], [2]	Categorical
	Inclusion of Vulnerable Groups in Outreach	CE6	[42], [17], [59], [60]	Categorical
	EV Evacuation Drills or Simulation Participation Rate	CE7	[61], [62], [63], [64]	Range (0–1)
	Feedback Mechanism from Residents on EV Emergency Planning	CE8	[65], [43], [66]	Boolean
Exposure Analysis	Hazard Risk Classification of the Region	EX1	[67], [68], [69]	Categorical
	Number of EV Owners in Hazard-Prone Zone	EX2	[70], [71]	Range (0–1)
	Availability of Charging Infrastructure in Exposed Zones	EX3	[72], [73], [74], [75]	Range (0–1)
	Data Integration of Exposure in Local Resilience Plans	EX4	[76], [77], [78], [79]	Boolean
	Regional Participation in Hazard Mapping or Simulation Studies	EX5	[80], [81], [82], [83]	Boolean
	Assessment of Critical EV Routes Crossing Hazard-Prone Zones	EX6	[72], [70], [84]	Categorical
Charging Infrastructure	Identification of EV chargers at greater risk of stress	CI1	[3], [2], [85]	Categorical
	Community awareness of charging energy needs for evacuation	CI2	[3], [2], [75], [84]	Categorical
	Assessment of public EV charging infrastructure for emergency evacuations	CI3	[7]	Categorical
	Emergency service vehicle (ESV) charging	CI4	[9], [86], [87]	Categorical
	EV charging management and allocation during emergencies	CI5	[8], [88], [89]	Categorical
	Protocols ensuring priority access for EV-based emergency vehicles	CI6	[53], [2]	Boolean
	Interoperability and standardization of EV charging	CI7	[4], [90], [91], [92], [93]	Categorical
	Real-time availability and status sharing protocols	CI8	[3], [94], [95], [96]	Boolean
	Spatial redundancy of charging network	CI9	[8], [97]	Range (0–1)
Mobile and Backup Power	Planning and deployment of mobile charging and backup power units during emergencies	MB1	[98], [99], [100], [101]	Categorical
	Large EV vehicles with V2X can be mobilized as mobile chargers	MB2	[98], [8], [4]	Boolean
	Critical EV charger locations are identified, and availability of backup power is confirmed	MB3	[102], [103]	Boolean
	Resilience of power systems at remote locations is maintained and dependable alternatives are suggested	MB4	[104], [105], [106]	Categorical
	Re-charging options for ZEVs with alternative fuels (e.g., hydrogen) are identified and available	MB5	[107]	Boolean
	Policies and incentives support public/private investment in backup power for EV charging infrastructure	MB6	[2], [108], [109]	Categorical
	Incentivize V2G charging for grid restoration post-disaster	MB7	[8], [25], [110]	Categorical
	Inter-agency coordination for mobile charging assets	MB8	[9], [111]	Categorical
Transportation and Evacuation Routing	Ensuring EV-Ready and Resilient Evacuation Routes	TR1	[3], [112]	Categorical
	Navigation apps used by the community have awareness about EV charging availability	TR2	[113], [114]	Boolean
	First responders have protocols in place to deal with EV fires and roadside assistance	TR3	[53], [115], [116], [117]	Boolean
	Separate evacuation routes for EVs with higher SoC/ICE cars	TR4	[4], [118], [3]	Categorical
	Shelters are distinguished based on the availability of EVCS	TR5	[2], [119], [120]	Boolean
	Availability of portable charging or swapping at strategic route points	TR6	[2], [121]	Boolean
	Inclusion of EV-specific evacuation scenarios in community drills	TR7	[122], [2]	Boolean

already operational when needed. During an active evacuation, engagement shifts toward real-time communication — such as EV-specific emergency alerts (CE1) — ensuring that pre-established plans are effectively executed. Post-event, community engagement supports recovery through feedback mechanisms (CE8) that inform iterative improvements to future evacuation plans. This temporal framing ensures that the burden of engagement does not fall on evacuees in the midst of a crisis, but rather on proactive preparation that empowers communities before disasters occur.

Community engagement is a foundational pillar of effective disaster preparedness. Numerous studies emphasize that empowering communities to participate in preparedness, response, and recovery processes enhances resilience, accelerates response time, and ensures more context-aware planning [65], [58], [57]. Engaged communities do not passively receive aid—they become co-creators of evacuation solutions that reflect their lived realities.

Moreover, the successful deployment of emerging technologies such as V2X depends heavily on proactive collaboration with entities like school districts, disability advocates, and first responders [123]. Such efforts require formal coordination mechanisms, including MOUs [44] and targeted awareness campaigns [45], to ensure readiness and resource sharing during crises. Pfefferbaum et al. [65] emphasize that participatory, assessment-based frameworks promote action-oriented resilience but also necessitate time-intensive coordination. Similarly, Norris et al. [58] argue that resilience is as much about collective capacity and adaptive social networks as it is about infrastructure, though measuring such capacities remains methodologically challenging. Collectively, these studies reveal that while community engagement is essential to the success of EV-focused evacuation frameworks, it must be strategically designed to account for operational complexity—such as coordination charging access, routing, and emergency roles—by insuring community responders are informed, and integrated into response systems.

Figure 5 provides a visual summary of the eight Community Engagement indicators and their abbreviations, illustrating how each contributes to EV evacuation readiness planning.

To evaluate community engagement in EV evacuations, the next section defines relevant indicators.

1) *EV User Communication and Readiness during Emergencies (CE1)*: This indicator evaluates whether real-time alerts, educational campaigns, and guidance specific to EV users are available and regularly updated [52]. Given the unique challenges EV drivers face during disasters—including limited range, long charging times, and power outages—clear, timely communication is essential for preparedness and safety [53].

Figure 4 illustrates an example of a multilingual wireless emergency alert issued by the FCC, demonstrating the type of accessible communication that CE1 aims to ensure for EV users during disasters.

2) *Emergency EV Charging Access Agreements and Public-Private Coordination (CE2)*: This indicator evaluates whether formal agreements and partnerships exist to secure EV charging access during emergencies through coordinated planning

Language	Template
American Sign Language (ASL)	Video link available at: https://youtu.be/0IP6m5y_Ww8
English	[SENDING AGENCY]: A 9-1-1 OUTAGE ALERT is in effect for [LOCATION]. Please contact police, fire, medical, or other emergency services directly at their local phone numbers in case of emergency. If you dial 9-1-1, you may not get help.
Arabic	[SENDING AGENCY]: (9-1-1) نسيب تكيبة اقتطاع خدمات الطوارئ [LOCATION] حتى [TIME]. يرجى الاتصال بالشرطة أو الإطفاء أو [LOCATION] في الخدمات الطبية أو غيرها من خدمات الطوارئ مباشرة على أرقام مراقبتهم المحلية في حالات الطوارئ. إذا اتصلت بالرقم 9-1-1، فقد لا تحصل على المساعدة.
Chinese (Simplified)	[SENDING AGENCY]: [LOCATION] 发布了 9-1-1 故障警报，当前生效中。如果发生紧急情况，请直接拨打当地电话号码联系警察、消防、医疗或其他紧急服务部门。如果您拨打 9-1-1，则可能无法获得帮助。
Chinese (Traditional)	[SENDING AGENCY]: [LOCATION] 發佈了 9-1-1 故障警報，當前生效中。如果發生緊急情況，請直接撥打當地電話號碼聯絡警察、消防、醫療或其他緊急服務部門。如果您撥打 9-1-1，您可能得不到幫助。
French	[SENDING AGENCY]: un AVIS D'INTERRUPTION DU 9-1-1 est en vigueur pour [LOCATION]. En cas d'urgence, veuillez contacter la police, les pompiers, les services médicaux ou tout autre service d'urgence directement en appelant leur numéro de téléphone local. Si vous composez le 9-1-1, vous risquez de ne pas obtenir d'aide.
German	[SENDING AGENCY]: Ein 9-1-1-AUSFALLALARM ist für [LOCATION] in Kraft. Bitte wenden Sie sich in Notfällen direkt an die Polizei, die Feuerwehr, die Rettungsdienste oder andere Notdienste unter den jeweiligen örtlichen

Figure 4: A multilingual wireless emergency alert guidance issued by FCC. [124]

and investment. This includes "Public-Private Partnerships (PPP)" structured models with clear roles and shared funding. For example, the Anqing project in China used a concession PPP agreement with performance-based subsidies to deploy widespread charging infrastructure, illustrating how contracts can ensure long-term investment [54]. Broader frameworks also highlight integrated planning platforms that align public incentives and private resources to support reliable, accessible charging services even in disaster contexts [55].

3) *Community Planning for EV-Based Emergency Evacuations (CE3)*: This indicator assesses whether the community has developed preparedness plans that address EV-specific constraints such as charging infrastructure, vehicle availability, and evacuation route prioritization [2]. Effective planning should consider strategies like adding DC fast chargers on evacuation corridors and integrating backup energy sources to mitigate grid outages [56]. It also involves evaluating infrastructure vulnerability, EV-to-charger ratios, and distance to evacuation centers using composite readiness scores to identify gaps and guide investments [18].

While evacuation routing indicators in the Transportation and Routing (TR) domain assess the operational and infrastructural readiness of evacuation corridors and charging availability, CE3 addresses a distinct and preceding dimension: whether the community, through its planning authorities, utility operators, and stakeholder organizations, has collectively developed and adopted preparedness plans that explicitly account for EV-specific constraints. This planning process, including stakeholder consultations, identification of infrastructure gaps, and coordination of evacuation strategies, is fundamentally a community engagement activity that must occur before operational routing decisions can be made. As such, CE3 serves as a prerequisite to the TR domain, ensuring that the social and institutional foundations for EV-based evacuation are established before infrastructure-level execution is assessed.

4) *Coordination among agencies to Conduct Emergency Response of EVs-Based Evacuation (CE4)*: This indicator determines whether local agencies coordinate with neighborhood organizations to test and improve their EV evacuation readiness through simulations and outreach [52]. This indicator also benefits from planning approaches that integrate social equity criteria into infrastructure deployment, supporting cross-sectoral coordination for equitable resource allocation during emergencies [36].

5) *Community Training and Awareness Programs (CE5)*: This indicator evaluates whether formal training or public education programs exist to prepare EV users for emergencies, including maintaining a high State-of-Charge (SoC) and planning charging along safe evacuation routes. This indicator evaluates whether formal training programs exist and how widely they are adopted, with an emphasis on educating EV users about pre-charging strategies and avoiding grid congestion—critical gaps identified in disaster preparedness [57], [58], [2].

Table III: Distinction between CE1 and CE5

Aspect	CE1	CE5
Focus	Real-time communication	Formal training programs
Timing	During emergency	Before emergency
Nature	Reactive	Proactive
Output	Alerts and guidance	Educated, prepared users
Audience	All EV users	EV users and responders

Although CE5 may appear similar to CE1, the two indicators address fundamentally different aspects of preparedness. CE1 focuses on real-time emergency communication — such as alerts and guidance issued *during* a disaster — while CE5 evaluates whether formal training and education programs exist to build EV users’ knowledge *before* a disaster occurs. For example, a community may have an excellent alert system (CE1), but if EV users have never been trained on pre-charging strategies or optimal State-of-Charge (SoC) targets, the alerts alone may be insufficient to ensure safe evacuation. Together, CE1 and CE5 form a complementary pair: one ensures real-time communication during emergencies, while the other ensures long-term preparedness through education and training. The similarity has been presented in Figure 18 and differences in Table III.

6) *Inclusion of Vulnerable Groups in Outreach (CE6)*: This indicator measures the extent to which outreach programs actively engage seniors, non-English speakers, people with disabilities, aligning with SDG equity principles and overlapping with E2 [42], [17]. Effective outreach requires intentionally identifying underserved communities that may be overlooked by conventional selection methods, as demonstrated by a multifaceted approach that combined demographic analysis with local interviews to uncover hidden or unorganized populations [59]. It also depends on culturally adapted, trust-building strategies that address barriers such as misinformation and institutional mistrust, shown to be critical in reaching low-income EV buyers in community-based campaigns [60].

7) *EV Evacuation Drills or Simulation Participation Rate (CE7)*: This indicator measures the extent to which communities conduct and participate in evacuation drills or sim-

ulations that include EV-specific constraints. Practice-based preparedness has been shown to improve real-world response and reduce confusion during emergencies [61]. Modern technologies such as agent-based simulation, digital twins, and GIS-enabled drill platforms—paired with new metrics like SoC-based evacuation completion rates and charger congestion modeling—enhance realism in preparedness exercises [63], while participatory models emphasize community-led simulation training to reveal local risks and strengthen planning through active engagement [64].

8) *Feedback Mechanism from Residents on EV Emergency Planning (CE8)*: This indicator evaluates whether local agencies have established channels—such as surveys, mobile applications, online tools, or community advisory boards—for residents to provide feedback on EV-specific evacuation planning. Such mechanisms support adaptive, iterative improvements by ensuring plans are informed by local needs and experiences [65], [43].

9) *Data Format for CE Indicators*: These indicators’ dataset can be collected in the form of categorical data in most of the cases. An agency can circulate a survey which can be filled out by a community. This community program participation (e.g. “Yes”, “No”, “In Progress”) can be captured using categorical data formats like CSV or spreadsheets. Moreover, indicators like CE2, can be captured in the form of boolean values.

Community Engagement (CE)

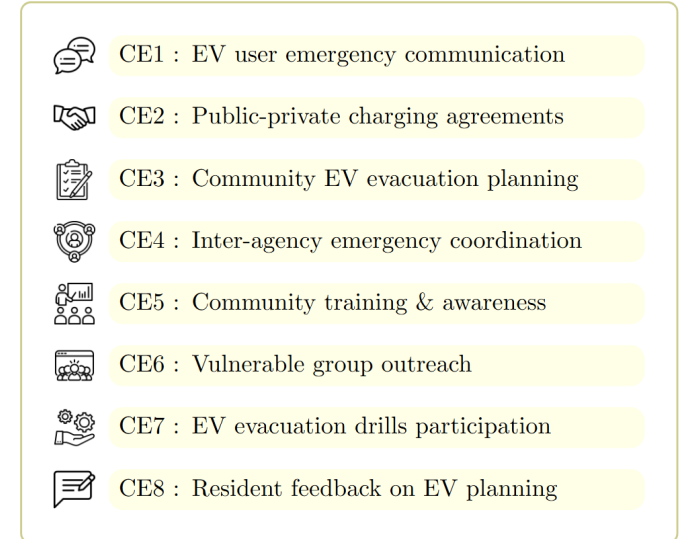


Figure 5: Visual representation of the community engagement domain, highlighting individual indicators and their abbreviations to facilitate understanding of their roles in EV evacuation planning.

B. Exposure Analysis

Exposure analysis plays a critical role in the pre-disaster stage of evacuation planning. It provides the foundation for understanding the vulnerability of a region to natural hazards, including floods, wildfires, earthquakes, storms among others.

Accurate and region-specific exposure assessments help communities, agencies, and planners identify who and what is at risk and how that risk translates into necessary preparedness efforts for EV evacuation.

Hazard Vulnerability Assessment (HVA) remains a cornerstone of disaster risk reduction, offering a systematic framework for evaluating potential disaster impacts and aligning available resources accordingly. In China, HVAs have demonstrated effectiveness in supporting region-specific preparedness and improving public health outcomes during emergencies [68]. Such assessments are especially pertinent for understanding how EV populations may be disproportionately affected in regions with limited access to charging infrastructure. Keller and Smith [72] extend this paradigm to enterprise and transportation resilience, advocating for the integration of geospatial hazard exposure data to inform mitigation strategies—principles that are directly applicable to EV evacuation planning in hazard-prone areas. Similarly, Inan et al. [73] and Raikes et al. [125] highlight the critical role of proactive risk assessments rooted in localized metrics such as hazard frequency and infrastructure vulnerability. FEMA [77] underscores the value of THVA (Target HVA) by offering structured tools—including likelihood-impact scoring matrices—to systematically identify site-specific vulnerabilities in transportation and power systems. While FEMA does not yet provide EV-specific planning indicators, their emphasis on risk-informed infrastructure assessments creates a foundational pathway for developing EV evacuation readiness frameworks. Complementary work by Zhong et al. [70] and Maurizio [80] further emphasizes integrating social vulnerability and environmental data to optimize pre-disaster resource allocation. The UNISDR [76] reinforces this approach through its national disaster risk assessment guidelines, which prioritize hazard mapping as an essential precursor to developing comprehensive and equitable resilience strategies—particularly for underserved or high-risk EV corridors.

Based on this extensive literature, following are the indicators proposed for measuring regional exposure and preparedness in EV evacuation contexts:

1) *Hazard Risk Classification of the Region (EX1)*: This indicator categorizes the region based on its historical and potential exposure to hazards such as floods, wildfires, hurricanes, or earthquakes. This indicator supports prioritizing evacuation infrastructure development in high-risk areas by using standardized, multi-hazard classification systems that ensure consistent and comparable assessments across regions [67], [68]. It also encourages more granular, spatially resolved risk mapping that accounts for local variability in hazard sources and vulnerabilities, improving the accuracy of zone-based planning and resource allocation [69].

Figure 6 illustrates an example of multi-hazard zone mapping for Santa Cruz County, encompassing wildfire, earthquake, tsunami, and flood risk areas, the type of spatial data that underpins indicator EX1.

2) *Number of EV Owners in Hazard-Prone Zones (EX2)*: This indicator quantifies the density of EV users in regions exposed to frequent disasters. This indicator helps prioritize resource allocation and evacuation alerts by identifying vul-

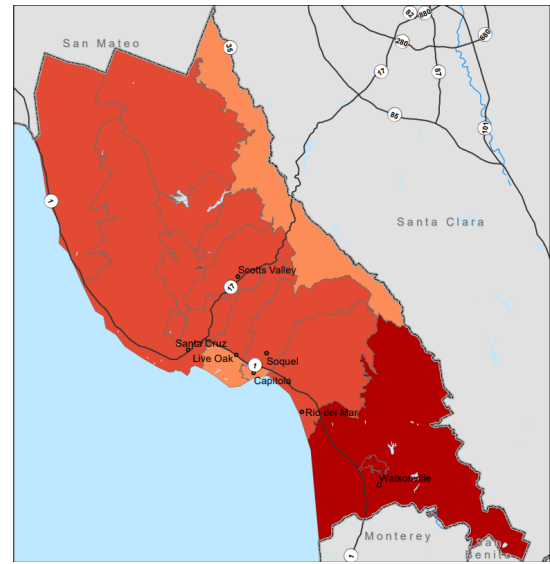


Figure 6: Map representing different hazard zones in Santa Cruz county that cater for wild fires, earthquakes, tsunamis and flooding. [18]

nerable sub-populations with specific mobility needs [70]. The approach relies on overlaying spatially resolved data on EV ownership with hazard-prone zones, following methodologies that have successfully combined population density grids and hazard maps to assess exposure at global scale [71].

3) *Availability of Charging Infrastructure in Exposed Zones (EX3)*: This indicator measures the number, type, and spatial distribution of EVCS in hazard-prone areas. Limited access in these zones can directly hinder evacuation by reducing recharging options for EV users during emergencies [72], [73]. Research highlights stark inequities in EVCS deployment, with lower-income and minority communities often lacking access to public and fast-charging infrastructure [74]. Moreover, the resilience of charging networks is critical: disruptions from hazards like hurricanes can severely limit service availability, underscoring the need for well-distributed charging infrastructure in vulnerable regions [75].

4) *Data Integration of Exposure in Local Resilience Plans (EX4)*: This indicator evaluates whether municipal or regional emergency plans incorporate exposure maps and hazard models (as suggested by UNISDR and FEMA) [76], [77]. This indicator emphasizes the need for plans to systematically integrate spatial data on hazard exposure to guide development and reduce vulnerability. Berke et al. [78] illustrate this by applying a scorecard method that assesses how well local plans align with mapped flood hazards to identify policy conflicts and gaps in mitigation. Yu et al. [79] further show that factors such as prior hazard experience significantly influence whether communities incorporate hazard exposure data into planning, underscoring the importance of institutional context for effective integration.

5) *Regional Participation in Hazard Mapping or Simulation Studies (EX5)*: This indicator measures whether the community or regional agencies have participated in any risk modeling studies or scenario planning exercises [80], [81].

This indicator emphasizes the importance of engaging local stakeholders in collaboratively assessing and preparing for risk. Tariq et al. [82] highlight the use of participatory systems thinking and system dynamics approaches to co-develop locally relevant hazard assessments and resilience indicators. Galang et al. [83] provide an empirical example of a participatory scenario planning workshop where regional agencies and organizations jointly envisioned future management strategies, strengthening systems thinking and trust among diverse actors.

6) *Assessment of Critical EV Routes Crossing Hazard-Prone Zones (EX6)*: This indicator identifies and evaluates major EV evacuation corridors intersecting with known flood plains or fire zones [72], [70]. This indicator emphasizes mapping and analyzing transportation routes that cross high-risk zones to ensure safe and reliable evacuation options for EV users. Zehra and Wong [84] highlight the vulnerability of transportation networks to wildfire damage, the need for redundancy in evacuation routes, and the importance of integrating hazard mapping with network planning to avoid bottlenecks and ensure accessibility during emergencies.

7) *Data Format for EX Indicators*: These indicators' dataset can be represented in the form of geospatial data. GIS shapefiles or geospatial databases for hazard risk mapping can include vector data (e.g. polygons for hazard zones) and raster data (e.g. elevation grids for flood risks) which can be helpful in presenting these indicators in the form of number and maps. Furthermore, numerical data representing the portions of EV owners in high-risk zones can be stored in a CSV or database format with columns for region, hazard zone and proportions.

Exposure Analysis (EX)

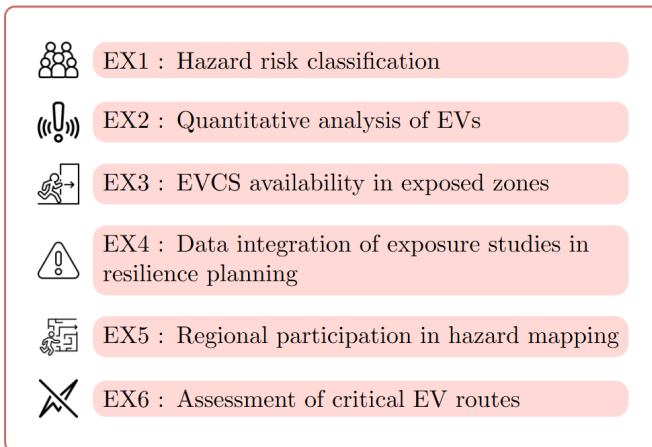


Figure 7: Visual representation of the Exposure Analysis domain, highlighting individual indicators and their abbreviations to facilitate understanding of their roles in EV evacuation planning.

Figure 7 summarizes the six Exposure Analysis indicators, enabling a quantitative and scalable approach to assess a region's EV infrastructure readiness relative to its natural hazard exposure.

C. Charging Infrastructure

The availability, accessibility, and robustness of EV charging infrastructure play a decisive role in the success of EV-based evacuation strategies. During emergency scenarios, especially those involving natural hazards, the functionality of EVCS is critical for both evacuees and emergency responders relying on electric fleets.

Charging Infrastructure (CI)

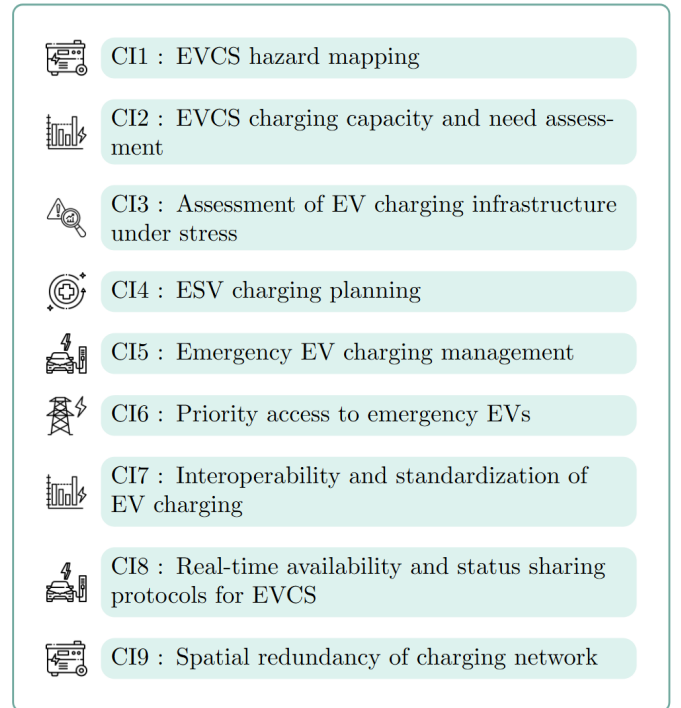


Figure 8: Visual representation of the charging infrastructure domain, highlighting individual indicators and their abbreviations to facilitate understanding of their roles in EV evacuation planning.

Babaei and Wong [2] underscore that power outages and increased demand can severely constrain the capacity of charging stations, necessitating spatially aware infrastructure assessments. Yusuf and Summers [3] further highlight the vulnerability of public EV chargers under surge conditions and advocate for their strategic placement in hazard-prone regions. Farzan et al. [4] emphasize the need for not only widespread charger availability but also interoperability between connector types (e.g., CHAdeMO, CCS), ensuring displaced users are not stranded due to incompatibility. Donaldson et al. [7] integrate this challenge into power system modeling, demonstrating that the absence of emergency power backups at key charging locations can lead to systemic failures along evacuation routes. In related work, Donaldson et al. [8] analyze load surges during wildfires and identify spatial redundancy—having multiple charging options within reach—as a vital design feature to prevent infrastructure bottlenecks. Hussain et al. [9] further argue for dynamic resource allocation during crises, including mobile chargers and energy storage at high-demand nodes,

which significantly improve response time and resilience of critical EV charging services.

Based on extensive literature review and research, the following indicators cover this domain completely, providing a credible metric to measure the readiness of the charging infrastructure of a region.

1) *Identification to EV chargers at greater risk of stress (CI1)*: This indicator measures charger availability in zones with high hazard exposure or predicted congestion [3], [2]. This indicator emphasizes the need to identify areas where charging demand may exceed available capacity during emergencies or peak travel periods. The NREL [85] report highlights the importance of forecasting corridor demand and anticipating future congestion hotspots along major travel routes.

2) *Community awareness of charging energy needs for evacuation (CI2)*: This indicator emphasizes the importance of public education on EV evacuation readiness, including SoC targets, estimated charging times, and local station maps. It is essential to consider advance communication on charging plans to reduce the risk of stations being overwhelmed during demand surges [3], [2] during hazards [75].

Table IV: Distinction between CE1, CE5 and CI2

Indicator	Focus	Domain
CE1	Real-time emergency alerts	Communication
CE5	General EV evacuation training	Education
CI2	Charging energy needs awareness	Infrastructure

Although CI2 may appear to overlap with CE1 and CE5, the three indicators address distinct aspects of preparedness. CE1 focuses on real-time emergency communication during a disaster, and CE5 evaluates formal training programs that build general EV evacuation knowledge before a disaster. CI2, in contrast, addresses a narrower and more technical dimension: whether the community has specific awareness of the *energy and charging requirements* needed to complete an evacuation, including estimated charging times, SoC targets, station locations along evacuation corridors, and the risk of station congestion during demand surges. While CE1 and CE5 operate at the level of general communication and education, CI2 is infrastructure-specific, it evaluates whether communities understand the charging demands that their EV fleet would place on the charging network during an emergency, which is a prerequisite for effective infrastructure planning and congestion management. These differences are better quantified in Table IV.

3) *Assessment of public EV charging infrastructure for emergency evacuations (CI3)*: This indicator includes geospatial audits and stress testing of publicly available chargers in evacuation zones. This indicator emphasizes evaluating whether charging infrastructure in evacuation areas can handle emergency demand surges. Donaldson et al. [7] demonstrate the use of geospatial modeling to identify critical network locations facing heightened stress from synchronized EV charging, and apply dynamic simulations that reveal significant pre-departure load spikes (caused by excessive vehicle being charged) and redistribution of power flows, supporting proactive planning and mitigation.

4) *Emergency Service Vehicle (ESV) charging (CI4)*: This indicator evaluates if infrastructure is prepared for larger vehicle classes (e.g., buses, rescue trucks) with higher charging demands [9]. This indicator emphasizes assessing whether local infrastructure planning accounts for the higher power and operational requirements of heavy-duty EVs. Methods developed by Rogge et al. [86] for joint optimization of bus fleet composition and charging station capacity to manage energy-intensive schedules effectively and Al-Hanahi's et al. [87] analysis of the challenges of deploying ultra-fast charging infrastructure for commercial and freight vehicles, including grid upgrades, corridor planning, and smart charging strategies can also be a helpful resource for planning of this indicator.

5) *EV charging management and allocation during emergencies (CI5)*: This indicator tracks local policies or systems that reallocate power/load to EVCS or coordinate access queues [8]. This indicator evaluates whether planning and operational strategies address the risk of congestion or grid stress during high-demand events. For example, Vutla and Chintham [88] highlight the role of optimal connector allocation and sizing to minimize average wait times and reduce grid impacts through advance planning. Meanwhile, Yang et al. [89] propose a dynamic management framework as presented in Figure 9 with real-time pricing and queue-based user choice models that adaptively redistribute demand across stations, demonstrating significant reductions in wait times and improved system resilience during peak conditions.

6) *Protocols ensuring priority access for EV-based emergency vehicles (CI6)*: This protocol examines if emergency fleets have designated, reserved, or expedited charging access, ensuring infrastructure can meet their critical needs during emergencies [53], [2]. Babaei and Wong highlight the need for planning charging station density and V2G readiness to support emergency EV fleets, while Adderly et al. emphasize policy frameworks ensuring sufficient fast-charging access along evacuation routes despite blackout risks and congestion.

7) *Interoperability and standardization of EV charging (CI7)*: This indicator confirms whether stations accept multiple charging standards to avoid user lockouts [4]. This indicator assesses if charging infrastructure supports diverse vehicles and technologies through adherence to common standards and protocols. Shin et al. [90] emphasize the importance of systematic compliance testing based on ISO/IEC 15118 and IEC 61850 to ensure charger-vehicle communication compatibility. Bohn et al. [91] demonstrate real-world testbeds that identify interoperability gaps across AC, DC, and wireless systems. Song et al. [92] review challenges in wireless EV charging alignment and efficiency, highlighting the need for unified standards. Kumar et al. [93] discuss the fragmentation of international charging standards and call for harmonized approaches.

Figure 10 illustrates the different connector types currently in use, highlighting the standardization challenges that CI7 seeks to address.

8) *Real-time availability and status sharing protocols (CI8)*: This indicator monitors if stations provide up-to-date operational data (e.g., wait time, outage) for route optimization [3]. This indicator assesses whether systems can reliably

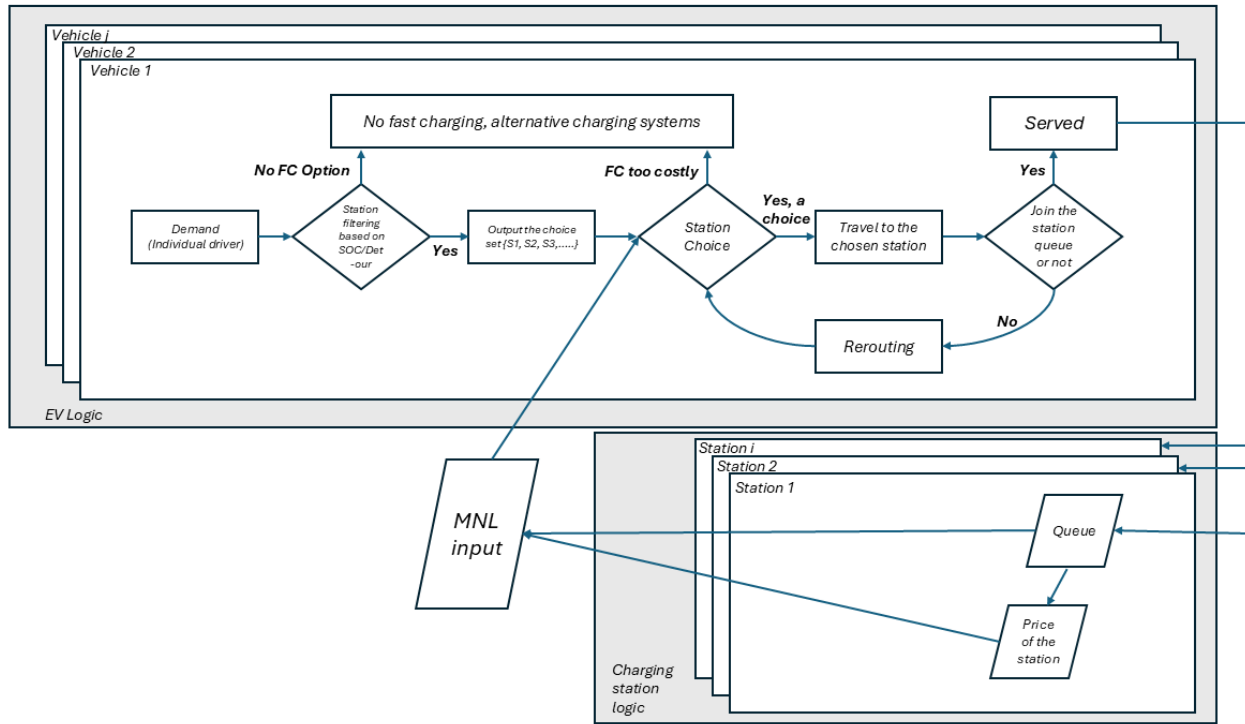


Figure 9: Framework extracted from [89] illustrating a dynamic framework with real-time pricing and queue-based user model.

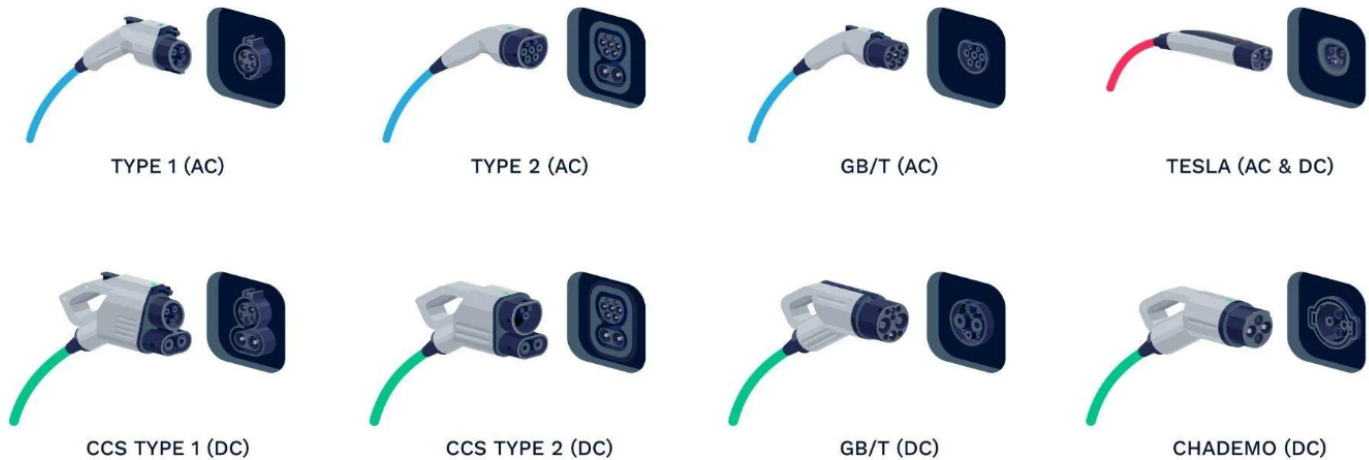


Figure 10: Different types of chargers used for charging [126].

share real-time information to support driver decision-making. Rinaldi et al. [94] highlight the need for sufficient mobile network capacity to handle the communication demands of real-time status updates. Elkashlan et al. [95] emphasize securing these exchanges against cyberattacks using intrusion detection to preserve data integrity. Chowdhury et al. [96] propose blockchain-based smart contracts to authenticate and safeguard operational data sharing across charging networks.

9) *Spatial redundancy of charging network (CI9)*: This indicator assesses whether hazard-prone areas have multiple chargers within short driving distances to avoid single-

point failure [8]. Yang et al. [97] highlight the importance of identifying underutilized "zombie stations" to reallocate resources efficiently, and propose optimizing station placement to ensure adequate coverage and reduce local congestion, directly supporting spatial redundancy in the network.

10) *Data Format for CI Indicators*: These indicators' dataset can be collected in the form of categorical data in most of the cases where data for whether charging stations are available and functional during emergencies (Yes/No or Good/Fair/Poor). Charging station locations and availability can also be represented using geospatial mapping. Range (n-

N) can also be used to define the availability of charging infrastructure in normal conditions as well as in emergencies.

Figure 8 summarizes the nine Charging Infrastructure indicators and their abbreviations, collectively covering the readiness of a region's EVCS for emergency scenarios.

D. Mobile and Backup Power

During large-scale evacuations or natural disasters, disruptions in the power grid can severely undermine the functionality of EV charging stations. The *Mobile and Backup Power* domain emphasizes the importance of resilient and flexible energy access in scenarios where fixed infrastructure is damaged, congested, or inaccessible. Across the literature, several complementary strategies emerge to address this vulnerability.

Multiple studies highlight the value of integrating mobile charging units (MCUs), backup generators, and distributed storage systems to maintain EV mobility during crises. For instance, [3] emphasize the deployment of MCUs in high-risk or off-grid areas to mitigate congestion and support emergency EV access when fixed stations are offline. Simulations from [7] demonstrate that the absence of backup systems can drastically reduce both evacuation capacity and emergency response effectiveness, highlighting the benefits of embedding distributed generation and storage in the charging network.

Figure 11 illustrates the contrast between normal and emergency system operations, showing how EVs, mobile backup sources, and communication networks dynamically interact to support grid resilience during disasters, a scenario directly addressed by the MB domain.

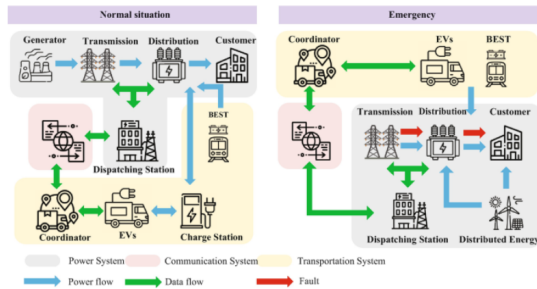


Figure 11: Illustration of system operations under normal and emergency conditions, highlighting how EVs, mobile backup sources, and communication networks dynamically support grid resilience and energy delivery during disasters.

In addition, [8] propose the use of V2G-enabled EVs to support microgrids and supply electricity to shelters, hospitals, or fire stations, particularly by repurposing heavy-duty EVs as mobile power units. The institutional dimension is also emphasized: [9] advocate for formal agreements between utilities, emergency services, and fleet owners to enable coordinated deployment and shared access to mobile energy assets during emergencies. Finally, [2] stress that infrastructure upgrades must be supported by policy incentives to drive investment in mobile charging systems and backup generation.

The following indicators were developed through a systematic review presented, to address the most critical aspects of mobile and backup power planning for EV-based evacuations.

While no finite set of indicators can claim to fully anticipate every possible natural hazard scenario or technological development, the proposed indicators are grounded in documented failures and lessons learned from a range of hazard types, including wildfires, hurricanes, floods, and grid outages, that represent the most common and severe threats to EV charging infrastructure. The indicators collectively address four critical dimensions of mobile and backup power resilience that recur across these hazard types: (1) the availability and deployment of mobile charging assets, (2) the integration of alternative and backup power sources, (3) the policy and institutional frameworks that enable coordinated deployment, and (4) the use of V2X technologies for grid restoration. Together, these dimensions provide a comprehensive but not exhaustive baseline for assessing a community's mobile and backup power readiness. As EV technology and disaster patterns evolve, additional indicators may be warranted, and the framework is intentionally designed to be modular and extensible to accommodate such updates.

1) *Planning and deployment of mobile charging and backup power units during emergencies (MB1)*: This indicator assesses utilities and agencies plans for rapid deployment of MCUs and generator trucks. Recent studies highlight the critical role of mobile energy assets in disaster preparedness. Saboori [98] shows that truck-mounted battery systems can be quickly deployed to restore electricity after grid disruptions. Weber [99] demonstrates the feasibility of hybrid mobile units powered by solar, batteries, and biodiesel to support backup power in remote locations. Yan [100] proposes an allocation method that optimizes the use of mobile and stationary storage units to ensure emergency coverage in urban areas. Guo [101] emphasizes the importance of geospatially informed scheduling to prioritize mobile energy delivery to isolated or high-need zones. Together, these contributions underscore the importance of flexible and well-coordinated deployment strategies for mobile power solutions.

2) *Large EV vehicles with V2X can be mobilized as mobile chargers (MB2)*: This indicator measures community capacity (the degree to which a region possesses the assets, agreements, and infrastructure needed to deploy large EVs as emergency power sources) to leverage electric buses or trucks to deliver power to shelters or evacuees. This indicator emphasizes the use of V2X technology in heavy-duty EVs (e.g., transit buses, truck-mounted BESS) as mobile energy sources during emergencies. Saboori [98] demonstrates how truck-mounted battery storage fleets can be dispatched to restore electricity in critical locations, increasing distribution network resilience. Donaldson et al. [8] highlight the feasibility of repurposing large EVs to support microgrids in disaster-affected zones. Farzan et al. [4] further underscore the need to plan for such deployments as part of broader evacuation strategies, especially in areas with vulnerable populations and limited fixed infrastructure.

3) *Critical EV charger locations are identified, and availability of backup power is confirmed (MB3)*: This indicator emphasizes the importance of ensuring resilience at key charging hubs by confirming the presence of backup power systems. Wang [102] highlights that safety and system continuity

in large-scale charging stations require redundant protection mechanisms and fail-safe electrical designs to mitigate the risk of sudden outages. In parallel, Powell and Johnson [103] demonstrate that the geographic placement and operational reliability of charging hubs significantly influence EV adoption, especially under adverse conditions, underscoring the need to strategically identify and reinforce critical stations with energy backup.

4) *Resilience of power systems at remote locations is maintained and dependable alternatives are suggested (MB4)*: This indicator tracks strategies to support rural or grid-isolated EVCS with solar, batteries, or MCUs. Recent reviews converge on the importance of decentralization and proactive planning for resilient energy delivery in remote areas. Bhusal [104] emphasizes the integration of system performance metrics with cross-sector coordination to prioritize recovery in vulnerable zones. Mahzarnia et al. [105] propose time-phased resilience actions and stress the value of microgrids and islanding for rural continuity. In contrast, Mohamed et al. [106] highlight the need for proactive investment in grid-hardening measures and geographic vulnerability mapping to ensure redundancy.

5) *Re-charging options for ZEVs with alternative fuels (e.g., hydrogen) are identified and available (MB5)*: This indicator evaluates the presence and planning of non-electric backup options such as hydrogen refueling stations (HRS) for use during emergencies. Isaac and Saha [107] highlight the importance of optimization models like the Flow-Refueling Location Model (FRLM) and multi-objective approaches to ensure the spatial availability of HRS, particularly in underserved or high-risk areas.

6) *Policies and incentives support public/private investment in backup power for EV charging infrastructure (MB6)*: This indicator assesses legislation and financial incentives encouraging backup infrastructure build-out [2]. Recent research emphasizes the importance of coordinated financial and policy frameworks to accelerate resilient infrastructure deployment. For instance, Rotimi-Ojo [108] highlights how the U.S. federal Bipartisan Infrastructure Law and Inflation Reduction Act incentivize public and private investment in critical EV infrastructure, including grid-resilient charging and alternative energy storage systems such as hydrogen and battery storage. In contrast, California's Assembly Bill 2127 mandates that the California Energy Commission (CEC) publish a biennial statewide assessment of electric vehicle charging infrastructure needs to support projected EV adoption—embedding a formal planning mechanism that aligns charger deployment with demand forecasts, equity goals, and grid capacity constraints [127]. According to the most recent projections under AB 2127, California will require over 1.01 million public and shared chargers by 2030 and 2.11 million by 2035, including both Level 2 and DC fast chargers [128]. To support this effort, the CEC has developed the EVSE Deployment and Grid Evaluation (EDGE) tool, which allows planners to model charger placement in relation to grid infrastructure. [129].

7) *Incentivize V2G charging for grid restoration post-disaster (MB7)*: This indicator analyzes community or utility programs rewarding EV users who contribute stored energy to grid recovery [8]. Recent studies have demonstrated the

potential of V2G strategies to support post-disaster power restoration in urban distribution networks. Du et al. [25] simulate multiple emergency scenarios and show that V2G-capable EVs can significantly accelerate service restoration if vehicles are optimally located and their SoC is sufficient. However, the study also emphasizes that without coordination or incentive mechanisms, the actual contribution from EVs remains limited. In comparison, Yang et al. [110] develop an agent-based response model that integrates EV user behavior and policy incentives. Their results highlight that effective incentives—such as financial compensation or priority charging—can increase EV user participation, enabling faster and more stable restoration across the grid.

8) *Inter-agency coordination for mobile charging assets (MB8)*: This indicator verifies whether local agencies, utilities, and private stakeholders have MOUs to coordinate the deployment of mobile energy assets during emergencies [9]. The *UCLA Luskin Center* report [111] highlights fragmented ownership of backup power systems in California and the lack of formal coordination mechanisms. It emphasizes the need for mutual aid agreements to enable rapid deployment, especially in underserved areas. For instance, in the San Diego region, an agreement between the local air district and the utility allowed mobile chargers and battery trailers to be pre-positioned at evacuation centers during wildfire season, improving response time during PSPS events. [130]

9) *Data Format for MB Indicators*: These indicators' dataset can be collected in the form of geospatial maps and categorical data, these datasets can pin point the locations for mobile and backup power generators to be deployed in case of an emergency. Time series data would also be helpful which could track real time changes in power system such as power outages and load changes during emergencies.

Figure 12 summarizes the eight Mobile and Backup Power indicators, illustrating how each contributes to maintaining EV mobility when fixed charging infrastructure is unavailable or compromised.

E. Transportation and Evacuation Routing

Transportation and routing during disaster evacuations represent a critical dimension of emergency preparedness, increasingly challenged by the growing presence of electric vehicles (EVs). Unlike internal combustion engine (ICE) vehicles, EVs introduce specific constraints—limited driving range, longer charging times, and dependence on vulnerable electrical infrastructure—that must be explicitly accounted for in evacuation planning.

A growing body of literature highlights the multifaceted challenges associated with integrating EVs into evacuation planning. Yusuf and Summers [3] identify that most current evacuation frameworks still reflect assumptions based on ICE vehicles, failing to accommodate EV-specific needs such as charger availability and SoC constraints. They emphasize that poor spatial alignment between EVCS and evacuation routes can cause congestion and premature battery depletion. Farzan et al. [4] complement this by proposing the integration of real-time EVCS availability into navigation platforms, advocating

Mobile & Backup Power (MB)

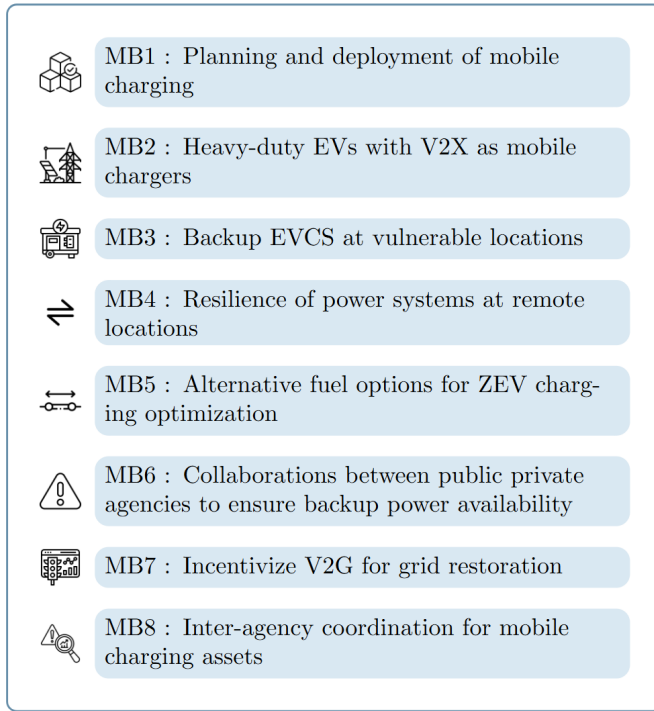


Figure 12: Visual representation of the mobile and backup power domain, highlighting individual indicators and their abbreviations to facilitate understanding of their roles in EV evacuation planning.

Transportation & Routing (TR)

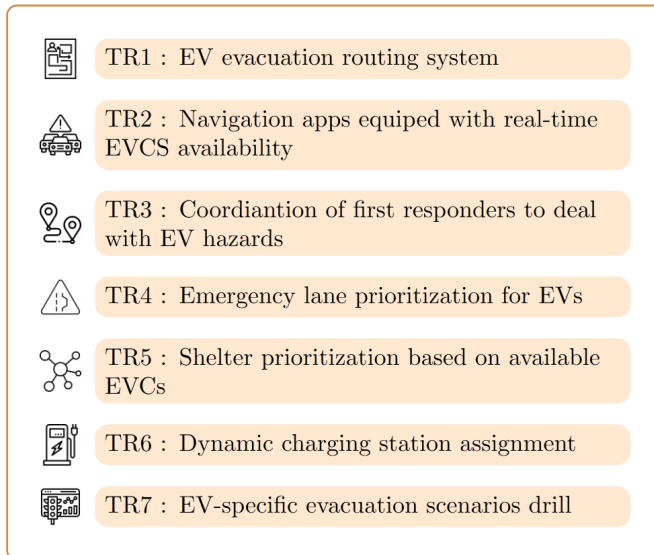


Figure 13: Visual representation of the transportation and routing domain, highlighting individual indicators and their abbreviations to facilitate understanding of their roles in EV evacuation planning.

for routing differentiation based on SoC levels to reduce bottlenecks and enhance equity. In parallel, Adderly et al. [53]

highlight a critical operational gap, noting the widespread lack of EV-specific training among first responders, which may hinder roadside assistance and compromise safety during EV-related incidents. Addressing infrastructural resilience, Donaldson et al. [7] recommend deploying mobile or trailer-mounted chargers along key evacuation corridors, demonstrating through simulation that such distributed assets improve system throughput and reduce reliance on fixed infrastructure. Peterson and Awwad [122] further argue that the exclusion of EV contingencies from current evacuation drills results in unrealistic and potentially ineffective response strategies. Finally, Babaei and Wong [2] propose a classification of emergency shelters based on EV charging capabilities, suggesting that such differentiation can guide evacuees more effectively and alleviate range anxiety during emergencies.

Based on the literature review presented, the following set of indicators are introduced that can define the domain completely but are not limited to only these:

1) *Ensuring EV-Ready and Resilient Evacuation Routes (TR1)*: This indicator measures whether critical evacuation paths are equipped with accessible charging infrastructure. While [3] emphasize that current evacuation planning rarely accounts for EV-specific needs such as charger access or SoC management along evacuation corridors, the *Long-Term Statewide EV Infrastructure Plan* developed by the Texas A&M Transportation Institute [112] further highlights the need to anticipate infrastructure deployment in vulnerable areas. Notably, the plan recommends evaluating the use of mobile charging trucks and off-grid charging systems during extreme weather events, such as hurricanes or wildfires, to improve emergency route resilience.

2) *Navigation apps used by the community have awareness about EV charging availability (TR2)*: This indicator assesses whether real-time charger data is integrated into routing platforms. This indicator is critical for enabling adaptive trip planning and reducing range anxiety among EV users. The mobile application *Enroute*, developed by Vijay et al. [113], as shown in Figure 14 offers a concrete example of a navigation tool that integrates real-time charger availability using crowd-sourced data and public APIs. It allows users not only to find and reserve nearby charging stations, but also to request on-demand mobile charging services when needed, enhancing confidence during long-distance travel.

In contrast, Hofer et al. [114] provide a broader analysis of 30 EV mobility apps in Europe and highlight that while many applications offer real-time charging data, few are truly integrated with routing engines or capable of dynamically adjusting routes based on charging needs. Compared to the high functionality of *Enroute*, most current platforms fall short of delivering a seamless EV-aware navigation experience.

3) *First responders have protocols in place to deal with EV fires and roadside assistance (TR3)*: This indicator reviews training, SOPs, and available response equipment for EV-related incidents [53]. Stave and Carlson [115] reveal that many firefighters in Swedish cities lack standardized protocols and rely on reactive learning to handle EV-specific risks like high-voltage exposure. Balcombe [116] highlights that U.S. fire departments, particularly smaller ones, often struggle to

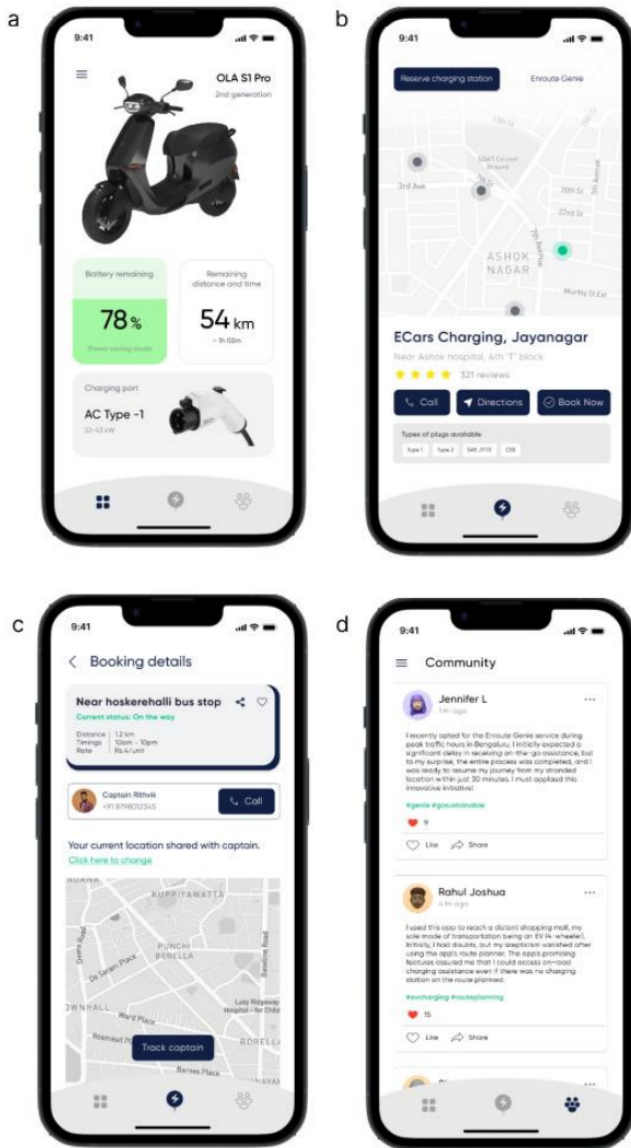


Figure 14: (a) Dashboard/home screen; (b) Nearby charging stations and selected charging station details; (c) Enroute Genie – Booking details; (d) Community screen. [113]

update SOPs in line with rapidly evolving EV technologies, calling for federal support and access to OEM data. Liu et al. [117] report that over 40% of U.S. first responders surveyed had never received EV-related training, underscoring the urgent need for national-level standardization in both training and equipment provisioning.

4) *Separate evacuation routes for EVs with higher SoC/ICE cars (TR4)*: This indicator evaluates if traffic routing differentiates vehicles to reduce congestion and charge anxiety [4]. The importance of incorporating EV-specific constraints such as SoC and charging capabilities is emphasized in recent literature. Li et al. [118] demonstrate through an optimization-based evacuation model that accounting for initial SoC and assigning EVs to different routes or charging stops based on their energy levels can significantly reduce evacuation time and grid stress. Similarly, Yusuf and Summers [3] highlight

that uniform evacuation plans overlook the heterogeneity in EV readiness, and they recommend differentiated strategies to manage low-SoC vehicles and charging infrastructure.

5) *Shelters are distinguished based on the availability of EVCS (TR5)*: This indicator tracks the number and location of shelters with Level 2 or DC fast charging facilities [2]. Although not focused directly on shelters, Kaya [119] demonstrates how multi-criteria GIS-based analysis can identify optimal EVCS locations by evaluating proximity to residential density, accessibility, and existing road infrastructure. Such methodology could be applied to prioritize shelters located in high-risk or high-demand zones. Complementing this, Ahmad et al. [120] show that nature-inspired optimization algorithms, such as PSO and GA, are effective tools to optimize charger placement in complex urban settings.

6) *Availability of portable charging or swapping at strategic route points (TR6)*: This indicator evaluates whether mobile charging solutions or battery swapping technologies are available along key evacuation corridors. Babaei et al. [2] emphasize the importance of deploying portable DC fast chargers and on-site storage to ensure continuity of EV mobility during emergencies, especially in areas where grid reliability is compromised. Their review highlights how mobile infrastructure can alleviate charging bottlenecks during evacuation scenarios. In contrast, Kandil et al. [121] examine battery swapping as a high-speed alternative, particularly effective for emergency fleets or dense urban zones. While both studies underline different technologies, they converge on the idea that fixed charging alone is insufficient under crisis conditions and that strategic deployment of portable or modular systems can enhance evacuation resilience.

7) *Inclusion of EV-specific evacuation scenarios in community drills (TR7)*: This indicator measures whether local planning includes table-top exercises or field drills featuring EV-specific contingencies [122]. While Babaei et al. [2] do not document existing drills, they emphasize a critical gap in current emergency planning: the widespread exclusion of EV-specific dynamics such as state-of-charge limitations, charging delays, and infrastructure dependencies. Their review calls for the development of integrated evacuation models that reflect EV behaviors and infrastructure constraints, and argue that without such inclusion, community drills risk remaining detached from real-world operational challenges.

8) *Data Format for TR Indicators*: These indicators' dataset can be collected in the form of categorical data and range data being represented on a geospatial map. This would help government agencies to identify roads to be specified for EV based evacuations only and deploy mobile chargers on only those roads.

F. Equity

As EVs continue to reshape disaster mobility, transportation planning must evolve to support their distinct characteristics. This domain (Figure 13) bridges operational logistics, infrastructure stress points, and behavioral routing strategies. The proposed indicators ensure that evacuation planning is not only inclusive of EVs but also optimized for their practical

limitations, ultimately contributing to a more resilient and adaptive emergency response system.

Equity is a crucial dimension in disaster preparedness, particularly for evacuations in socio-technically diverse communities. Social equity means ensuring fair distribution of resources and opportunities, so that disaster response does not leave marginalized groups more vulnerable [42].

In the context of EV evacuations, equity encompasses not only access to infrastructure and affordability but also inclusive communication and meaningful participation in planning. Numerous studies underscore the challenges and opportunities in embedding equity within EV resilience strategies. Lemphers et al. [131] and Coleman et al. [132] point to political, economic, and infrastructural barriers that disproportionately affect low-income and minority communities, making equitable electrification a complex but essential goal. These barriers—whether stemming from exclusionary policymaking, financial limitations, or lack of physical infrastructure—intersect to undermine equitable EV evacuation preparedness, often leaving the most at-risk populations least able to evacuate safely and efficiently. Research on blackout scenarios [133], [134] further emphasizes that without targeted policies, electrification may exacerbate mobility gaps during crises. To address this, scholars have proposed the integration of socio-demographic variables in infrastructure modeling [135], [136], although such approaches are often data-intensive and require model calibrations to reflect unique characteristics, demographics and infrastructure conditions of a specific geographic area. Broader critiques [46], [43] argue that equity remains peripheral in many resilience frameworks and call for a shift toward justice-based, inclusive planning. Real-world failures, such as the unequal impacts observed during Winter Storm Uri [47], revealed how low-income and minority communities faced longer outages, lacked access to backup heating, and were cut off from critical services due to inadequate infrastructure and communication breakdowns. These failures affirm that EV evacuation planning must actively confront systemic disparities—including access to resilient mobility and energy systems—to ensure equitable outcomes in the transition to cleaner transportation.

Figure 15 summarizes the six Equity indicators and their abbreviations, collectively ensuring that EV evacuation planning addresses the needs of vulnerable and underserved populations across all disaster phases.

1) Emergency Support Programs for Low-Income EV Owners (EQ1): This indicator evaluates whether dedicated subsidies, incentives, or support mechanisms are in place to help low-income EV owners charge, evacuate, or repair their vehicles during disasters [123]. Past incentive programs like California’s EFMP (Enhanced Fleet Modernization Program) have demonstrated that low-income households are highly responsive to financial support—offering up to \$9,500 for vehicle replacement in disadvantaged communities [31]. Additionally, structural disparities in EVSE—such as charger deserts in low-income areas, limited access in multifamily housing, and profit-driven siting practices—continue to exacerbate vulnerability during emergencies [32].

Equity (EQ)

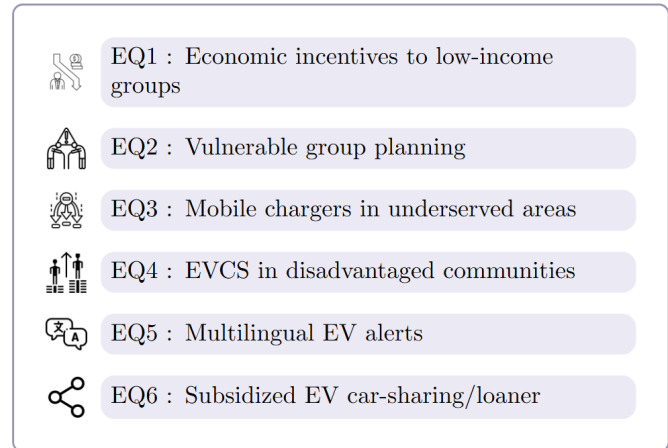


Figure 15: Visual representation of the equity domain, highlighting individual indicators and their abbreviations to facilitate understanding of their roles in EV evacuation planning.

2) Participation of Vulnerable Groups in Planning (EQ2): This indicator measures the inclusion of disabled, elderly, and socioeconomically disadvantaged individuals in emergency planning processes, community drills, and stakeholder engagement activities [17]. Studies have shown that participatory methods are essential to uncover hidden vulnerabilities and enhance preparedness [33], [34]. Post-disaster analyses from Hurricane Katrina revealed that elderly, disabled, and low-income individuals were often left behind due to inaccessible transportation, communication failures, and a lack of representation in decision-making [35]. These failures underscore the critical need for proactive, inclusive engagement in planning and drills to ensure that preparedness measures reflect the needs of all community members.

3) Mobile Charger Deployment in Underserved Areas (EQ3): This indicator assesses whether mobile charging units are pre-positioned or deployable in rural, remote, or underserved communities where charging infrastructure is sparse [3]. Recent studies highlight the role of mobile charging systems in closing infrastructure gaps in low-income neighborhoods [36], [37], where fixed installations remain scarce due to structural and spatial constraints.

4) EVCS Access in Disadvantaged Communities (EQ4): This indicator quantifies the percentage of public EV charging stations located within or near low-income or historically underserved neighborhoods. This indicator evaluates whether charging infrastructure deployment mitigates or reinforces documented spatial inequities. Research has consistently demonstrated significant disparities in access to public EV charging stations across income lines, with fewer stations per capita in low-income neighborhoods [38], [39]. These disparities are not only spatial but also include issues of reliability and functionality in disadvantaged areas [40]. City-level analyses further confirm such patterns, highlighting the need for local, equity-focused planning to address access gaps [41]. Including this metric helps identify underserved

areas and supports targeted investments to promote equitable transportation electrification. As shown in Figure 16, Santa Cruz exhibits significantly lower charging intensity compared to San Francisco, despite being only 80–90 miles apart. Figure 17 further illustrates the stark disparity in EVCS deployment between the two regions, underscoring the equity gaps that indicator EQ4 is designed to capture.

5) *Access to EV Alerts and Multilingual Communication (EQ5)*: This indicator evaluates the availability of real-time alerts, evacuation instructions, and charging information in multiple languages and accessible formats. This indicator measures whether communication strategies address the needs of non-English speakers, individuals with disabilities, and other at-risk groups [42], [43]. Effective emergency communication requires multilingual, culturally adapted, and accessible formats to ensure no population is left behind during evacuations [44]. Research also emphasizes the importance of trust-building, training for public information officers, and the integration of social media to reach diverse audiences with timely and accurate information [45]. [124] presents a multilingual wireless emergency alert issued by the Federal Communications Commission (FCC) involving the alert being issued in English, Arabic, Chinese, French, German, Haitian Creole, Hindi, Italian, Korean, Portuguese, Russian, Spanish, Tagalog, Vietnamese.

6) *Subsidized Access to EV Car-Sharing or Loaner Programs (EQ6)*: This indicator evaluates whether local governments or private sector partners provide affordable, temporary access to EVs through car-sharing or ride-share programs for individuals without private EV ownership who need to evacuate. It considers publicly funded or subsidized models designed to improve equitable access for low-income and underserved populations [46], [47]. Community-based programs such as CRuSE and SiLVERS have demonstrated the potential of tailored pricing and multilingual apps to serve rural and senior riders, while also facing barriers such as insurance costs and technology access [48]. Publicly funded initiatives like BlueLA and Miocar can expand access in low-income neighborhoods but often depend on sustained funding and strong partnerships [49]. Siting subsidized EV car-share stations near affordable housing has shown promise in addressing transit gaps [50], and addressing user concerns over vehicle availability and range anxiety is essential to encourage participation [51].

7) *Data Format for EQ Indicators*: These indicators' dataset can be collected in the form of categorical data in most of the cases as it would highlight the availability of EV chargers and ease of transportation for subsidized communities. In order to better the representation of this dataset, it can be presented on a map resulting on a geospatial dataset being developed integrated with categorical and range data signifying availability of EV chargers (where and how many) in certain percentage of equitable community living in a region.

IV. DISCUSSION

The proposed multi-domain indicator framework addresses a significant gap in EV-based evacuation planning by introduc-

ing a structured approach that spans all disaster phases: pre-event, during-event, and post-event. While prior studies have explored technical and infrastructural aspects of resilience [9], [2], few have translated these into actionable policies or socially grounded strategies. For instance, indicators such as charging infrastructure interoperability or V2G-enabled mobile charging have been technically documented [7], [4], but remain largely unimplemented at scale. Additionally, no standardized methodology has been proposed to identify which parameters should be prioritized when designing EV evacuation plans. This section analyzes these gaps, explores overlaps across indicators, aligns each domain with disaster phases, and discusses the framework's strengths and limitations alongside recommendations for future refinement.

Comparative Gap Analysis

Despite increasing awareness, several indicators remain underdeveloped or lack operational uptake. For example, although EV fire safety is gaining attention, many jurisdictions have not adopted formal protocols for first responders dealing with EV roadside incidents [53]. This highlights a substantial training gap: new SOPs are needed to equip first responders with the skills to safely and effectively manage EV-specific emergencies (Indicators EQ2, EQ3, EQ5, CE1, CE3, CE5, CE7).

Similarly, mobile charging infrastructure, though critical for evacuation resilience [7], has yet to be standardized in most regions. While states like Florida and Michigan have experimented with mobile charger deployment during evacuations [137], these efforts have typically relied on reactive, real-time data rather than pre-established strategic plans. Establishing protocols for charger pre-positioning based on hazard exposure and traffic modeling could enhance efficiency and reduce delays in future emergencies.

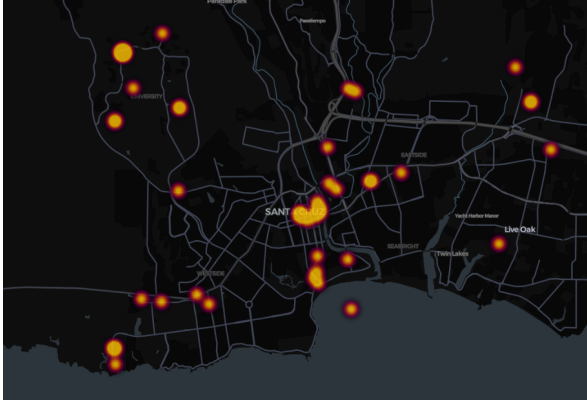
Equity-related indicators—such as charger siting in disadvantaged communities or improving accessibility for people with disabilities—are frequently mentioned in policy and planning documents [132], [43] but are rarely operationalized through simulation drills or design requirements. This disconnect between social vulnerability assessments and emergency operations represents a major implementation gap.

In addition, the literature lacks discussion around differentiated evacuation routing for EVs and ICE vehicles. Indicator TR4 addresses this explicitly, proposing separate traffic strategies to reduce congestion and improve efficiency. Similarly, the exposure domain remains underexplored in EV-specific evacuation contexts. This study provides a structured indicator set that considers not only real-time operations but also hazard exposure of routes, regions, and shelter infrastructure—elements critical to designing context-responsive and inclusive evacuation systems.

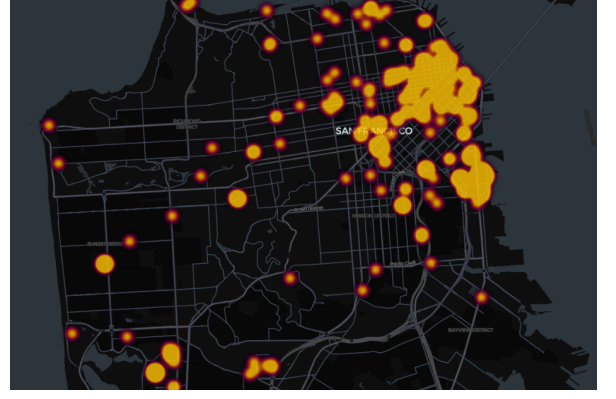
Temporal Alignment of Domains

Each of the six proposed domains aligns with specific phases of disaster response:

- Pre-event (Equity, Exposure Analysis, Community Engagement, Charging Infrastructure Planning, Mobile and

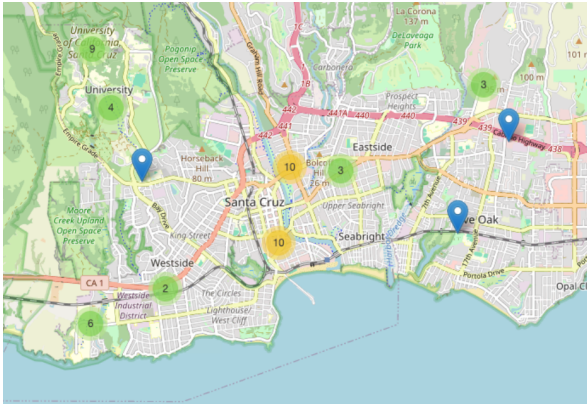


(a) Santa Cruz charging intensity heatmap

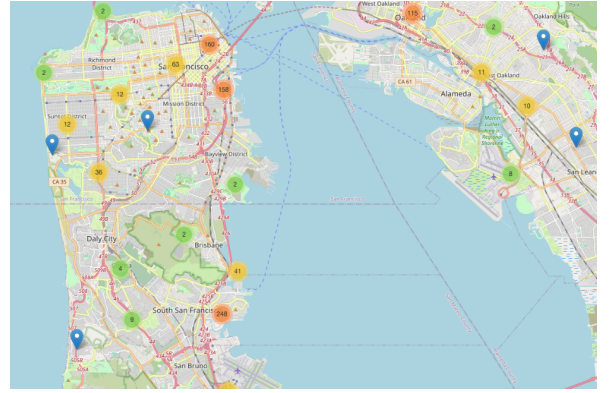


(b) San Francisco charging intensity heatmap

Figure 16: Visualization of charger usage intensity at two different regions in California.



(a) Santa Cruz EVCS distribution



(b) San Francisco EVCS distribution

Figure 17: EVCS deployment in both regions to match charging demand intensities.

Backup Power Planning, Transportation and Routing Planning): These domains focus on long-term planning, social inclusion, and hazard awareness. They support vulnerability assessments and proactive preparedness measures [68], [57].

- During-event (Charging Infrastructure, Mobile and Backup Power, Transportation and Routing): These domains address real-time operational demands such as charging availability, route optimization, and emergency logistics [4], [3]. For example, during a wildfire, the power system may experience surges in demand from evacuees, requiring rapid response mechanisms and mobile charging stations to manage infrastructure stress and maintain mobility.
- Post-event (Mobile and Backup Power, Community Engagement): These domains also support recovery operations, including power restoration via V2G (MB7), post-disaster return flows, and community feedback loops (CE8) [9]. These processes are often sequential, beginning with grid restoration and extending to transportation re-entry and procedural evaluation.

These are presented in tabular format in Table V, where each indicator is presented with their contributions in each of the temporal stage.

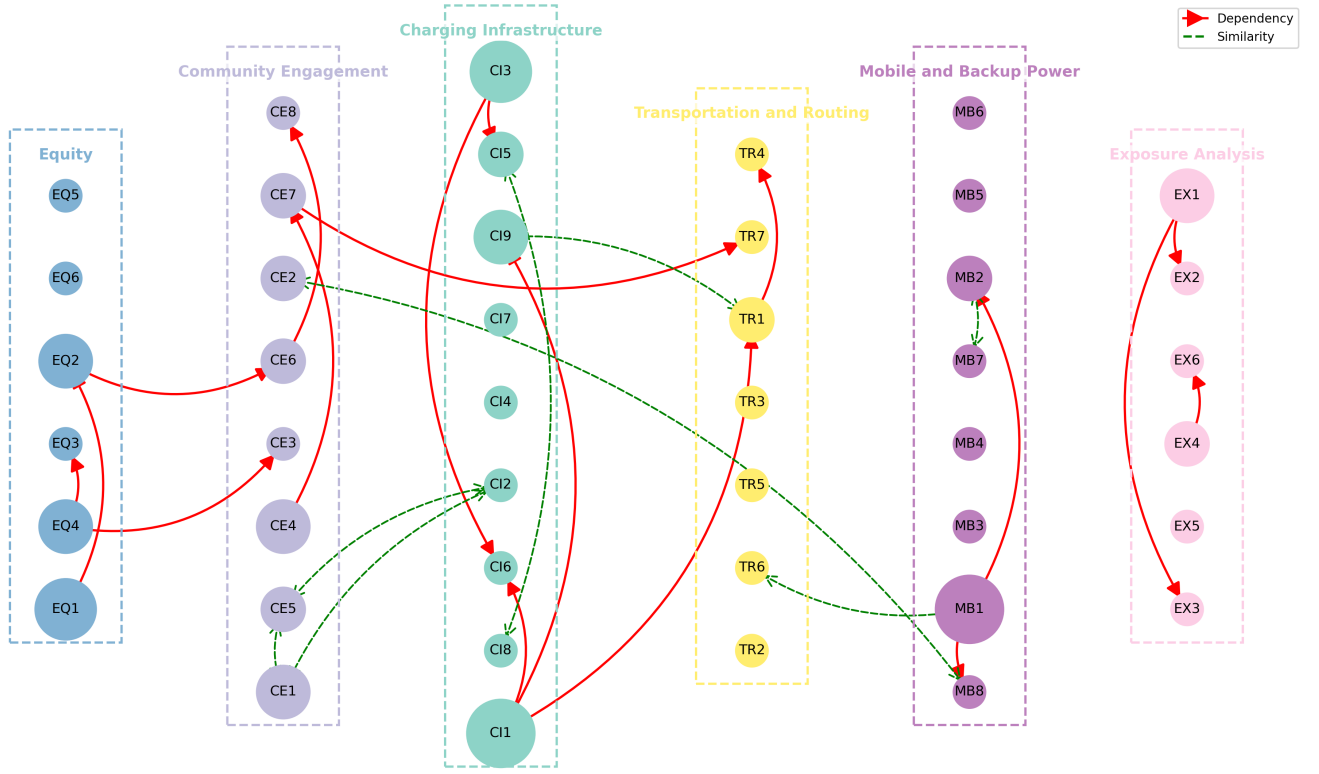
Framework Overlap and Strategic Integration

To analyze the relationships between indicators, a structured framework was developed to distinguish between logical dependencies (explicit links where one indicator relies on another) and semantic similarities (conceptual overlaps based on shared attributes). Each indicator was characterized according to five descriptive dimensions:

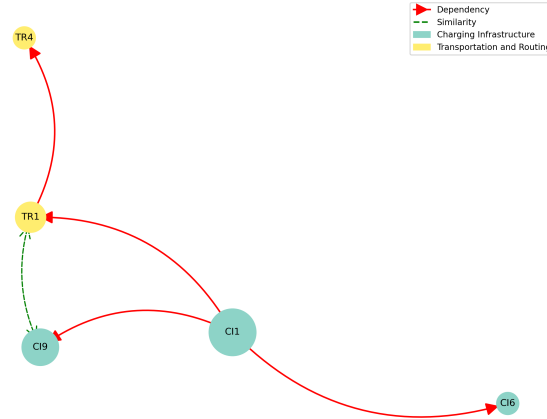
- Domain
- Disaster phase
- Action category
- Stakeholder category
- Dependencies

Each indicator was then encoded as a set of categorical attributes derived from these dimensions. For example, an indicator may belong to a given domain (e.g., Charging Infrastructure), be associated with a specific disaster phase (e.g., preparedness), target a type of action (e.g., planning or deployment), involve specific stakeholders (e.g., local authorities or utilities), and include explicit dependencies on other indicators.

Pairs of indicators were compared using the Jaccard similarity coefficient, a standard metric for assessing similarity between categorical sets [138]. For two indicators A and B ,



(a) Visualization of cross-domain similarities and interdependencies among resilience indicators.



(b) Visualization of the strongest interdependencies among resilience indicators.

Figure 18: Visual representations highlighting the interconnected and cross-domain nature of the proposed EV evacuation resilience framework.

the similarity is defined as:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

where $|A \cap B|$ represents the number of shared attributes and $|A \cup B|$ the total number of distinct attributes across both indicators. In practice, since each indicator is described by five dimensions, similarity scores can be interpreted as ratios (e.g., $3/5$ indicates that three out of five attributes are shared). A value of $J(A, B) = 1$ indicates complete overlap, while $J(A, B) = 0$ indicates no shared characteristics.

Dependencies were incorporated in two ways: (i) as explicit directed links between indicators, and (ii) as an additional

similarity factor when two indicators rely on a common third indicator. This allows the framework to capture both structural relationships and thematic proximity.

The analysis shows that the highest similarity scores are generally observed within the same domain. For instance, $J(CE1, CE5) = 3/5$ and $J(TR7, CE7) = 3/5$ reflect strong overlaps in community preparedness and engagement activities.

However, several cross-domain relationships were also identified. For example, $J(CE1, CI2) = 2/5$, $J(CE5, CI2) = 2/5$, and $J(CE2, MB8) = 2/5$ highlight the interdependence between communication strategies, charging infrastructure, and mobile backup planning. Within the infrastructure domain, similarities

Table V: Temporal Alignment of Indicators Across Disaster Phases

Domain	Ind.	Disaster Phase	Role in Disaster Phase
Equity (EQ)	EQ1	Pre-event	Establishes financial support programs for low-income EV owners before disasters occur
	EQ2	Pre-event	Ensures vulnerable groups are included in planning processes before emergencies
	EQ3	Pre-event/During-event	Pre-positions mobile chargers in underserved areas; deployable during evacuation
	EQ4	Pre-event	Assesses EVCS coverage gaps in disadvantaged communities to guide investment
	EQ5	Pre-event/During-event	Develops multilingual alert systems before disasters; activated during emergencies
	EQ6	Pre-event/During-event	Establishes subsidized EV car-sharing programs available during evacuations
Exposure (EX)	EX1	Pre-event	Classifies regional hazard risk to prioritize EV infrastructure investment
	EX2	Pre-event	Maps EV owner density in hazard zones to guide pre-disaster resource allocation
	EX3	Pre-event	Assesses EVCS availability in exposed zones before disaster strikes
	EX4	Pre-event	Integrates hazard exposure data into local resilience plans
	EX5	Pre-event	Engages communities in hazard mapping and simulation exercises
	EX6	Pre-event	Identifies critical EV evacuation routes crossing hazard-prone zones
Community Engagement (CE)	CE1	Pre-event/During-event	Develops real-time EV alert systems before disasters; activated during emergencies
	CE2	Pre-event	Establishes public-private charging agreements before emergencies occur
	CE3	Pre-event	Develops community EV evacuation plans before disasters
	CE4	Pre-event	Coordinates inter-agency EV evacuation protocols before emergencies
	CE5	Pre-event	Trains EV users on pre-charging strategies and evacuation route planning
	CE6	Pre-event	Engages vulnerable groups in outreach and planning before disasters
	CE7	Pre-event	Conducts EV-specific evacuation drills to build community readiness
	CE8	Post-event	Collects community feedback to improve future EV evacuation plans
Charging Infrastructure (CI)	CI1	Pre-event/During-event	Identifies high-stress chargers before disasters; monitored during evacuations
	CI2	Pre-event/During-event	Educates communities on charging needs before disasters; critical during surge
	CI3	Pre-event/During-event	Audits public EVCS capacity before disasters; stress-tested during evacuations
	CI4	Pre-event/During-event	Plans ESV charging infrastructure before disasters; activated during response
	CI5	During-event	Manages and allocates EV charging resources during active evacuations
	CI6	During-event	Ensures priority charging access for emergency vehicles during evacuations
	CI7	Pre-event	Ensures charger interoperability standards are in place before disasters
	CI8	During-event	Shares real-time charger status and availability during active evacuations
	CI9	Pre-event/During-event	Ensures spatial redundancy of charging network before and during disasters
Mobile & Backup Power (MB)	MB1	Pre-event/During-event	Plans MCU deployment before disasters; executed during active evacuations
	MB2	During-event	Mobilizes V2X-capable EVs as emergency power sources during evacuations
	MB3	Pre-event/During-event	Identifies critical charger locations and confirms backup power availability
	MB4	Pre-event/During-event	Plans remote EVCS resilience strategies; activated during grid outages
	MB5	Pre-event/During-event	Identifies hydrogen refueling options before disasters; accessed during response
	MB6	Pre-event/Post-event	Establishes policies and incentives for backup power investment
	MB7	Post-event	Incentivizes V2G participation to restore grid after disasters
	MB8	Pre-event/During-event	Establishes inter-agency MCU coordination agreements before disasters
Transportation & Routing (TR)	TR1	Pre-event/During-event	Plans EV-ready evacuation routes before disasters; used during evacuations
	TR2	During-event	Ensures navigation apps display real-time EVCS availability during evacuations
	TR3	Pre-event/During-event	Trains first responders for EV incidents before disasters; deployed during response
	TR4	During-event	Differentiates EV and ICE evacuation routes during active evacuations
	TR5	Pre-event/During-event	Designates EVCS-equipped shelters before disasters; directs evacuees during response
	TR6	Pre-event/During-event	Pre-positions portable chargers along corridors; deployed during evacuations
	TR7	Pre-event	Includes EV-specific scenarios in community drills before disasters

such as $J(CI5, CI8) = 2/5$ and the stronger relationship $J(CI9, TR1) = 4/5$ emphasize the link between network redundancy and evacuation route resilience. Similarly, in the mobile power domain, pairs such as $J(MB1, TR6) = 2/5$ and $J(MB7, MB2) = 2/5$ illustrate operational connections between mobile charging, battery swapping, and vehicle-to-grid strategies.

Overall, this analysis provides two key insights. First, it confirms that indicators tend to cluster within domains, supporting the internal consistency of the framework. Second, it reveals critical cross-domain interactions that would not be captured by a purely hierarchical structure. These results are particularly useful for identifying redundancies, prioritizing indicators with high centrality, and ensuring that interdependent aspects (e.g., infrastructure, communication, and routing) are jointly considered in evacuation planning.

A complete overview of identified dependencies and similarity ratios is provided in the supplementary materials as an Excel table.

Figure 18a synthesizes these relationships: dotted arrows indicate semantic similarity, while red arrows represent explicit dependencies. The node size reflects indicator centrality, with larger nodes corresponding to indicators that influence a greater number of others. The strongest interdependencies are detailed separately in Figure 18b.

Comparison with Existing Models

Compared to existing frameworks—such as the DOE’s EV Grid Integration Initiative or the EU Green Deal roadmap—this approach places greater emphasis on social metrics and cross-sectoral coordination. While traditional models often prioritize emissions reduction or grid stability, our framework includes social equity (EQ1–EQ4), community-based governance (CE7–CE8), and flexible energy deployment (MB1–MB8) as core pillars.

It also builds upon previous indicator schemes used for scoring in [18], offering a more adaptable, regionally-tailored structure for local agencies, national labs, and NGOs.

Collection of Datasets for Indicators

Collection of the required datasets for all 36 indicators can prove to be a significant challenge for government agencies, especially in regions with limited resources and infrastructure. However, a practical approach is proposed to collect these datasets that can be scaled based on the available resources and priorities

- **Use of existing data sources:** Many of the indicators can be derived from already existing data sources, including public records, utility data, and disaster response reports.

For example, charging infrastructure data can often be obtained from local electric utility providers, while community engagement indicators may rely on data from local government outreach programs. Data on exposure analysis can be sourced from hazard mapping studies conducted by agencies like FEMA or local disaster management authorities

- **Collaboration with stakeholders:** Collecting some of the more complex or granular data will require collaboration across multiple stakeholders. This may include local authorities, utility companies, transportation agencies, and community organizations. For example, data on mobile and backup power could be shared by energy utilities or emergency response organizations that maintain backup power systems.
- **Prioritizing based on region:** Recognizing the diverse nature of disaster preparedness across regions, it is suggested that government agencies prioritize the collection of a subset of the 36 indicators based on local needs and disaster risks. In regions prone to wildfires, for example, exposure analysis and charging logistics indicators may be of higher priority. Conversely, in urban areas with extensive infrastructure, transportation routing and community engagement may take precedence.
- **Pilot programs and scalable implementation:** It is proposed that smaller-scale pilot studies be conducted to evaluate specific indicators in high-risk areas. These pilots can help assess the practicality of data collection methods and demonstrate the framework's value before scaling up to include all 36 indicators. Agencies can gradually expand the scope of data collection as they gain more experience and resources.

Some of the data sources have been identified in Table VI. This is not an exhaustive list, but have been developed keeping US states in perspective, specifically California.

Weighted Scoring via MCDA

The adaptability of this framework is wonderful. Although detailed discussion on weighted scoring is part of future work and out of scope of this paper but this framework can be used to develop pre-disaster temporal stage scoring and the same framework can be used for during disaster temporal stage and take steps at the moment.

When agencies are developing planning stages of a region to prepare it for EV-based evacuation, higher priorities can be given to preparatory indicators i.e. Exposure Analysis (EX), Community Engagement (CE) and Equity (EQ). Once these datasets are developed, other domains can be optimized based on these datasets. One example weighting strategy for preparatory purposes can be to give 20% to EX, CE and EQ and then prioritizing TR and CI by giving it 15% weight and 10% to MB.

In case of a disaster during the disaster stage, a new scoring methodology can be developed just for that specific temporal stage which can focus only on the evacuation and the resilience of the power system under stress. In such case, weighting strategy can be to give 30% weight to TR, 20% to MB, 20% to CI and then 10% each to EQ, CE and EX.

Scope and Limitations

This framework is grounded in peer-reviewed literature, government reports, and practitioner insights. However, emerging technologies (e.g., hydrogen refueling, SoC prediction algorithms) and evolving disaster patterns (e.g., heat domes, climate migration) may necessitate the development of additional indicators or redefinition of existing ones. Moreover, as EV technology develops more, it would change the weighting strategy of the indicators and domains altogether, if the charging time are significantly reduced, then the weighting of Mobile and Backup Power would be reduced and more weightage can be given to domains like Transportation and Evacuations Routing as then the main focus would be to prevent any blockage on the evacuation route. Such assumptions can be made for other indicators as well in case of other technological advancements.

Furthermore, the present analysis primarily addresses household- and community-level resilience. Broader systemic considerations, such as EV supply chain vulnerabilities, energy workforce readiness, and fleet-level logistics, are beyond the current scope but offer promising directions for future research and policy development.

This framework also lacks ability to capture the uncertainty of human behavior in case of emergency and life or death situations but has introduced indicators to best prepare the community. It has been proven that humans tend to hoard up supplies and equipments when there is a hazard. It tends to happen in a situation where there is no estimate of how long this situation would last. As seen during Covid-19 pandemic, grocery stores were out of number of basic necessities because people were not sure how long the pandemic would last. A similar outcome can be expected in case of an evacuation using EVs. In a talk with community service department's official at Merced, CA, it was stated that based on official's experience, evacuees of a wild fire would tend to fill their cars with their belongings rather than helping other evacuees escape the danger. No model or framework can cater for these behaviors but can introduce domains like Community Engagement and indicators like CE1, CE3, CE4, CE5, CE6 and CE8 to best prepare communities for such scenarios.

V. OPPORTUNITIES AND STRATEGIES

The proposed indicator framework not only reveals critical gaps in EV evacuation readiness but also surfaces unique opportunities for actionable strategies at the intersection of mobility, equity, and resilience. These indicators can serve as operational levers for federal, state, and municipal stakeholders to enhance preparedness, response, and recovery phases of disaster management. This sections helps identifying ways of using the framework to revolutionize EV evacuation.

Indicator-Informed Strategic Planning

Several indicators overlap across domains as shown in Figure 18a—such as community awareness (CE2, CI5), charging resilience (CI1, CI3, TR6), and equitable access (EQ3, CE6)—highlighting the need for integrated planning across sectors. By cross-referencing indicators, agencies can develop

Table VI: Domains and its Indicators with Data Sources

Domain	Ind.	Disaster Phase	Data Sources
Equity (EQ)	EQ1	Pre-event	U.S. Census Bureau ACS; DOE Low-Income Energy Affordability Data (LEAD); LIHEAP program data; State EV incentive databases
	EQ2	Pre-event	CDC Social Vulnerability Index (SVI); FEMA Vulnerable Populations data; Local emergency management plans
	EQ3	Pre-event/During-event	DOE AFDC EVCS locator; Census block-level income data; Local utility grid maps; Emergency logistics databases
	EQ4	Pre-event	NREL EVI-Pro data; EPA EJScreen; CalEnviroScreen; DOE AFDC station database
	EQ5	Pre-event/During-event	U.S. Census Language Use data; FEMA IPAWS; Local OES multilingual communication plans
	EQ6	Pre-event/During-event	Shared mobility operator data (Lyft/Zipcar); State EV program registries; FEMA Individual Assistance records
Exposure (EX)	EX1	Pre-event	FEMA National Risk Index; USGS Hazard maps; NOAA climate risk data; State OES hazard mitigation plans
	EX2	Pre-event	State DMV EV registration data; FEMA flood/fire zone shapefiles; U.S. Census TIGER shapefiles; NREL EV adoption models
	EX3	Pre-event	DOE AFDC station locator API; FEMA hazard zone GIS layers; PlugShare/ChargePoint operator APIs
	EX4	Pre-event	State multi-hazard mitigation plans; FEMA BRIC grant data; Local general plans; CalOES resilience frameworks
	EX5	Pre-event	FEMA Community Preparedness data; Local emergency exercise after-action reports; Hazard simulation studies
	EX6	Pre-event	State DOT evacuation route maps; FEMA hazard zone shapefiles; Google/HERE road network APIs; CAL FIRE perimeter data
Community Engagement (CE)	CE1	Pre-event/During-event	FEMA IPAWS; Wireless Emergency Alert (WEA) system; State OES alert platforms; Utility outage notification systems
	CE2	Pre-event	State PUC filings; Utility-EV operator MOUs; ChargePoint/Electrify America partnership data; DOE Clean Cities Coalition
	CE3	Pre-event	Local emergency operations plans (EOPs); FEMA CPG 101 guidance; AAA EV survey data; State evacuation plan repositories
	CE4	Pre-event	State OES inter-agency coordination records; FEMA NIMS data; ESF-1 Transportation Annexes; MOU databases
	CE5	Pre-event	State DMV EV training records; DOE Clean Cities outreach data; Local utility EV education program records
	CE6	Pre-event	CDC SVI; FEMA disability integration data; Local CBO/NGO partnership records; HUD CDBG data
	CE7	Pre-event	Local OES drill after-action reports; FEMA exercise evaluation data; State fire agency drill records
	CE8	Post-event	Post-disaster community surveys; FEMA after-action reports (AARs); Local OES debrief records
Charging Infrastructure (CI)	CI1	Pre-event/During-event	ChargePoint/EVgo/Blink real-time APIs; Utility SCADA systems; NREL station utilization reports
	CI2	Pre-event/During-event	DOE EV consumer education materials; Utility demand-response program records; State PUC consumer data
	CI3	Pre-event/During-event	DOE AFDC station database; State PUC EVCS inspection records; Operator uptime/reliability reports
	CI4	Pre-event/During-event	FEMA ESF-13 records; State emergency vehicle fleet databases; Utility infrastructure plans
	CI5	During-event	Real-time ChargePoint/EVgo APIs; Utility demand response systems; State OES resource tracking systems
	CI6	During-event	State emergency vehicle registry; Utility priority customer lists; FEMA ESF-1/13 operational data
	CI7	Pre-event	SAE International standards database; DOE interoperability testing records; CharIN compliance data
	CI8	During-event	PlugShare/ChargePoint live APIs; State 511 evacuation systems; FEMA situational awareness platforms
	CI9	Pre-event/During-event	NREL EVI-RoadTrip tool; DOE AFDC network gap analysis; State energy resilience plans; GIS coverage analysis
Mobile & Backup Power (MB)	MB1	Pre-event/During-event	State OES resource inventories; DOE mobile charging unit registries; Utility emergency response plans
	MB2	During-event	State V2G pilot program records; EPRI V2X deployment data; OEM (Ford/Nissan) V2H/V2G capability databases
	MB3	Pre-event/During-event	Utility backup power infrastructure maps; DOE critical facility databases; State energy emergency plans
	MB4	Pre-event/During-event	NREL microgrid database; FEMA hazard mitigation plans; Utility distribution system resilience reports
	MB5	Pre-event/During-event	DOE AFDC hydrogen station locator; FCHCA station data; CARB FCEV registry
	MB6	Pre-event/Post-event	State energy commission policy databases; CPUC/PUC dockets; FEMA BRIC/HMGP grant records
	MB7	Post-event	State V2G enrollment records; Utility demand response databases; EPRI V2G pilot study data
	MB8	Pre-event/During-event	State OES inter-agency MOUs; FEMA ESF-12 Energy Annex; Utility mutual aid agreement records
Transportation & Routing (TR)	TR1	Pre-event/During-event	State DOT evacuation route databases; FEMA hazard zone GIS; Google Maps/HERE APIs; NREL EV range modeling
	TR2	During-event	PlugShare/ChargePoint live APIs; Apple Maps/Google Maps EV routing data; State 511 system feeds
	TR3	Pre-event/During-event	State fire/EMS training records; NFPA EV incident databases; DOE alternative fuel vehicle responder guides
	TR4	During-event	State DOT contraflow/route management systems; FEMA evacuation route GIS; Traffic management center data
	TR5	Pre-event/During-event	FEMA National Shelter System (NSS); Red Cross shelter databases; State OES shelter plans; DOE AFDC locator
	TR6	Pre-event/During-event	State OES resource staging plans; DOE mobile charging asset inventories; FEMA logistics system data
	TR7	Pre-event	Local OES drill records; FEMA HSEEP data; State fire agency drill after-action reports

a streamlined strategic framework, supported by a decision matrix or dashboard tool as explained in [18]. This would allow prioritization of interventions (e.g., deploying mobile charging units in socioeconomically vulnerable zones [137], [2]) and strategic investments based on hazard exposure maps [68], [73].

Opportunities for Temporal Integration

Our indicator framework can be segmented into phases of disaster management as explained in the previous section but they can be used to develop a strategy as follows:

- **Preparedness:** Community engagement (CE5–CE8), training protocols for responders (TR3), and pre-disaster hazard vulnerability assessments (EX1–EX3) are essential in advance of emergencies [77], [65]. These would prepare the first-responders and evacuees for any kind of hazard.
- **Response:** Indicators such as real-time charger availability (CI8), mobile charging access (TR6), and EVCS prioritization protocols (CI6) become operationally critical during active evacuation [4], [9].
- **Recovery:** Feedback loops (CE8), infrastructure performance evaluations (CI3, EX5), and iterative community input support continuous improvement and resilience-building post-disaster [3], [18].

This temporal segmentation ensures resource allocation and policy focus are aligned with the evolving needs of EV users and infrastructure resilience. The preparedness, response and recovery are not only limited to the set of indicators introduced, but can be extended based on the need of time and technological advancements.

Policy and Simulation Integration

Indicators such as TR7 (inclusion of EVs in drills) and CE7 (participation rate in simulations) underscore a unique opportunity: the design of holistic simulation environments that account for EV-specific routing, charging congestion, and equity challenges. State pilot programs (e.g., There is a Florida’s deployment of mobile chargers [137]) offer promising blueprints for expanding EV-inclusive policy trials. Further, agencies can adopt these indicators into platforms like the National Risk Index or state-level evacuation apps. After further research these can be used to develop a resilient and stable evacuation plan for EVs.

Interoperability and Standardization Strategies

Many indicators point to fragmented or siloed data practices (charging status APIs, EVCS maps, or exposure datasets) are not consistently accessible or interoperable. Strategies should include adopting open standards for charger status reporting (CI8), enforcing interoperability (CI7), and harmonizing vulnerability indices with infrastructure planning (EQ4, EX2) [72], [139]. These align with Sustainable Development Goals calling for inclusive, digital-ready resilience frameworks [12].

Advancing Cross-Domain Collaboration

Finally, the convergence of mobility, power systems, social equity, and digital infrastructure creates new grounds for interdisciplinary innovation. By fostering collaborative networks between departments of transportation, energy utilities, emergency management agencies, and community organizations, indicators like CE6 (vulnerable group inclusion) and CI4 (ESV charging) can guide development of holistic evacuation ecosystems [61], [43].

In summary, these indicators offer more than assessment: they serve as a blueprint for coordinated EV evacuation readiness strategy, actionable across spatial scales and temporal stages. The next step involves deploying these indicators in pilot regions, evaluating performance during drills, and scaling successful strategies through collaborative data sharing and federal guidance.

VI. CONCLUSION

This paper presents a novel and comprehensive framework of 36 resilience indicators to assess and enhance EV emergency preparedness. With the accelerated adoption of electric vehicles and the increased frequency of climate-induced disasters, conventional evacuation strategies (primarily designed for ICE vehicles) need to be designed. EVs introduce unique operational dependencies such as charging availability, battery state-of-charge, and grid reliance, which must be systematically incorporated into resilience planning. Yet, the literature to date has lacked a structured and actionable methodology for integrating EVs into disaster response ecosystems.

The proposed framework addresses this critical gap by categorizing resilience into six interrelated domains: *Equity, Exposure Analysis, Community Engagement, Charging Infrastructure, Mobile and Backup Power, and Transportation and Routing*. Each domain is supported by literature-driven indicators that are not only theoretically grounded but also designed for practical implementation by planners, municipalities, and emergency management authorities.

The framework’s key strength lies in its temporal alignment with the disaster management life cycle, segregating indicators across the **preparedness**, **response**, and **recovery** phases. For instance, equity and community engagement indicators help establish inclusive pre-disaster protocols, while charging logistics and mobile power indicators become pivotal during real-time evacuation. Post-event strategies benefit from feedback loops, infrastructure audits, and V2G-supported grid restoration.

Furthermore, this framework is designed to be both modular and extensible. It acknowledges indicator overlaps, enabling cross-domain coordination and integration into decision-support tools such as simulation environments or regional readiness dashboards. Importantly, it also emphasizes interoperability—of infrastructure, datasets, and institutions—recognizing that fragmented systems significantly undermine real-world resilience.

By integrating insights from technical literature, policy reviews, national and international resilience frameworks (e.g., SDGs, UNDRR), and simulation-based models, this paper

translates theory into actionable tools for EV-based evacuation planning. It underscores the need for a shift from reactive policy development to proactive, equity-centered, and community-informed resilience strategies.

In essence, the framework not only serves as a diagnostic tool to assess current gaps but also as a roadmap for designing forward-looking, inclusive, and scalable EV evacuation systems. As EV adoption continues to surge and climate risks intensify, this work provides a timely and essential foundation for transforming disaster resilience paradigms to meet the needs of a decarbonized, electrified, and more equitable future.

Future Work: This study opens several avenues for future research. These include: (1) applying the indicator framework to real-world case studies and pilot regions; (2) developing weighted scoring mechanisms using multi-criteria decision analysis (MCDA); (3) integrating simulation-based scenario testing with dynamic resilience metrics; and (4) refining indicators for alternative ZEV types (e.g., hydrogen fuel cell vehicles) or rural/tribal communities.

The authors envision that this framework will evolve into a standardized planning tool to be adopted by transportation agencies, emergency planners, utilities, and researchers committed to climate adaptation and sustainable mobility.

Note: During the preparation of this work the author(s) used ChatGPT (version 5.1) to assist with text refinement and language polishing. After using this tool/service, the author(s) reviewed and edited the content as needed and take full responsibility for the content of the published article.

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